Available online at www.sciencedirect.com

ScienceDirect

J. Differential Equations 266 (2019) 1073–1120

Journal of Differential Equations

www.elsevier.com/locate/jde

Existence and stability of spatially localized patterns

Elizabeth Makrides *, Björn Sandstede

Division of Applied Mathematics, Brown University, Providence, RI 02912, USA

Received 9 February 2018; revised 5 July 2018
Available online 3 August 2018

Abstract

Spatially localized patterns have been observed in numerous physical contexts, and their bifurcation diagrams often exhibit similar snaking behavior: symmetric solution branches, connected by bifurcating asymmetric solution branches, wind back and forth in an appropriate parameter. Previous papers have addressed existence of such solutions; here we address their stability, taking the necessary first step of unifying existence and uniqueness proofs for symmetric and asymmetric solutions. We then show that, under appropriate assumptions, temporal eigenvalues of the front and back underlying a localized solution are added with multiplicity in the right half plane. In a companion paper, we analyze the behavior of eigenvalues at $\lambda = 0$ and inside the essential spectrum. Our results show that localized snaking solutions are stable if, and only if, the underlying fronts and backs are stable: unlike localized non-oscillatory solutions, no interaction eigenvalues are present. We use the planar Swift–Hohenberg system to illustrate our results.
© 2018 Elsevier Inc. All rights reserved.

Keywords: Homoclinic snaking; Planar patterns; Partial differential equations; Stability; Evans function; Swift–Hohenberg equation

1. Introduction

In this paper, we are interested in spatially localized structures. Specifically, we will investigate interested localized structures that consist of an oscillatory pattern with finite spatial range that appears within a homogeneous background state. These structures can be viewed as gluing a front (connecting a background state at $x = -\infty$ with a spatially periodic structure at $x = \infty$) and

* Corresponding author.
E-mail addresses: elizabeth_makrides@brown.edu (E. Makrides), bjorn_sandstede@brown.edu (B. Sandstede).

https://doi.org/10.1016/j.jde.2018.07.064
0022-0396/© 2018 Elsevier Inc. All rights reserved.

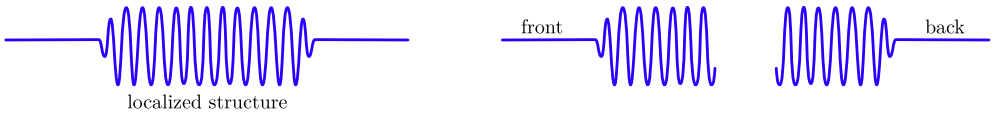


Fig. 1. Shown are a spatial localized structure (left), a front (center) and a back (right). The horizontal axis is space, while the vertical axis represents the dependent variable (such as crime intensity, displacement, or vegetation growth).

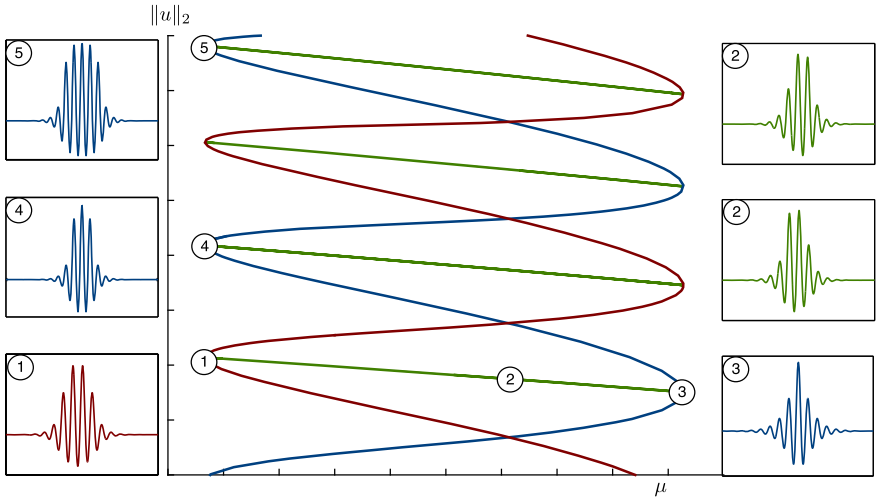


Fig. 2. The center shows the bifurcation diagram of localized structures as a function of a systems parameter μ. The insets show the spatial profiles of the localized solutions as the branch is traversed. Note that there are two unbounded branches (red, blue) corresponding to symmetric profiles that are connected by horizontal curve segments (green) corresponding to asymmetric profiles. Solutions along the red branch have a minimum in the center and solutions along the blue branch have a maximum in the center; their dynamical behavior is the same. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

a back (connecting a spatially periodic structure at $x = \infty$ with a background state at $x = -\infty$) together to create a localized structure; see Fig. 1 for an illustration. As systems parameters are varied, fronts and backs typically appear and disappear in saddle-node bifurcations. It is then of interest to understand the bifurcation structure of the accompanying localized structures: interestingly, these diagrams often exhibit snaking as illustrated in Fig. 2. Localized structures and snaking diagrams appear in a wide variety of physical contexts, including crime hotspots [33,18], plane Couette flow [32], optical cavity solitons [23], cellular buckling [12], and vegetation patterns [24]. Additional physical settings that support localized structures include liquid crystals, magnetoconvection, current filaments in a driven semiconductor-gas discharge system [8,27,2,5,6]. The review articles [16,10,17] contain an even larger selection of examples and references.

In this paper, we provide a unified approach to rigorously address the stability of symmetric and asymmetric localized snaking solutions. Our starting assumption is the existence of fronts and backs: the goal is then to show that these can indeed be glued together to create localized structures, to trace out the resulting bifurcation diagram if the existence regions of fronts and backs are known, and to determine the stability properties of the localized structures assuming the stability of fronts and backs is known. Previously, in [4], an analytic gluing argument was used to show the existence of symmetric solutions, while a condition was given for the continuation of branches of asymmetric solutions appearing at pitchfork bifurcations, and a formal

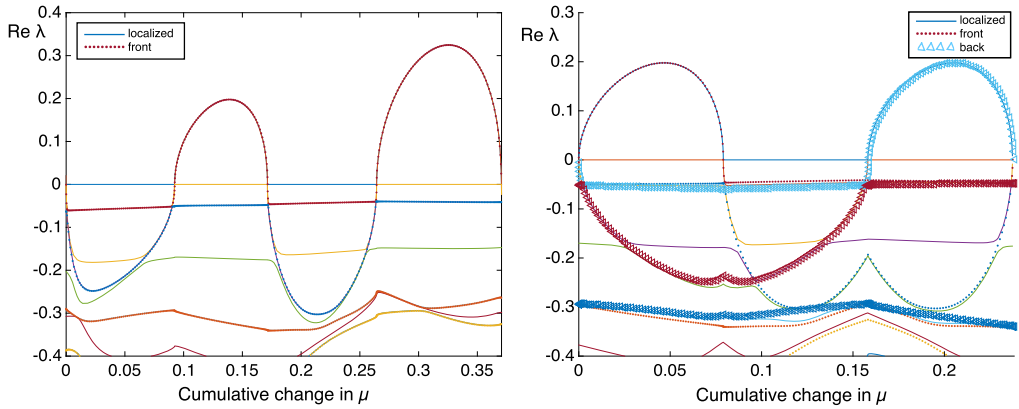


Fig. 3. Left: Eigenvalues for symmetric localized solutions (solid), with eigenvalues for corresponding fronts (dots). Right: Eigenvalues for asymmetric localized solutions (solid), with eigenvalues for corresponding fronts (dots) and backs (triangles) overlaid. See §7 for additional details.

argument based on collective coordinates was suggested to predict the stability of asymmetric solutions. Here we rigorously show that both symmetric and asymmetric localized solutions are exponentially close in the length L of the patterned region to the underlying fronts and backs, which allows us to prove that in the right half plane, eigenvalues of the localized solution are exponentially close in L to those of the front and back, and are added with multiplicity.

When combined with [22], which includes a proof that if the front and back have simple eigenvalues at $\lambda = 0$ in an appropriate sense, the eigenvalue of the localized solution at $\lambda = 0$ will also be simple: this shows that localized snaking solutions are stable if, and only if, the underlying fronts and backs are stable. This is in contrast to the case for localized solutions constructed from gluing together fronts and backs connecting stable rest points, where a second interaction eigenvalue arises near $\lambda = 0$ [1,25,28]. The intuition behind this result comes from Fig. 1: fronts and backs that connect to spatially periodic states can be glued together only when the phases of the periodic orbits match. In contrast, fronts and backs that connect to homogeneous rest states can, to leading order, be glued together at arbitrary distances as their asymptotic profiles agree up to exponentially small corrections: the additional freedom of shifting the profiles arbitrarily against each other creates a second interaction eigenvalue that is not present in the case where the asymptotic profiles are periodic.

Our desire to understand the stability of localized solutions is partially motivated by the numerically computed spectra shown in Fig. 3, which illustrates typical eigenvalue behavior along a snaking branch of symmetric solutions and a bifurcating branch of asymmetric solutions. This sort of behavior has been seen previously in [7] and [3]: the computations indicate that the saddle-node and pitchfork eigenvalues can be tracked into the right-half plane as they move into the essential spectrum. This is at odds with the theoretical results in [13,14], which suggest that, upon reaching the essential spectrum, the eigenvalues should generically become invisible (without going into too much detail, as the eigenvalues cross the origin, they should move onto the wrong Riemann sheet of an appropriate Evans function that is analytic in $\gamma = \sqrt{\lambda}$ and therefore correspond to a resonance pole rather than genuine eigenvalue). As mentioned above, in [22], we will use the results developed here to construct an extended Evans function that explains the unexpected behavior of the saddle-node and pitchfork eigenvalues as they cross the origin $\lambda = 0$.

1.1. A motivating example

One particular, and in some sense canonical, example of a PDE exhibiting localized snaking patterns is the Swift–Hohenberg equation:

$$U_t = -(1+\Delta)^2 U - \mu U + \nu U^2 - U^3, \tag{1.1}$$

with $U = U(x,t)$ or $U(x,y,t)$ and $x \in \mathbb{R}$ or $(x,y) \in \mathbb{R}^2$. The above equation is referred to as the quadratic-cubic form, due to the powers in the nonlinearity, while

$$U_t = -(1+\Delta)^2 U - \mu U + \nu U^3 - U^5, \tag{1.2}$$

is referred to as the cubic-quintic form. As our results are applicable to either form of the nonlinearity, we will use a general nonlinearity $g(U,\nu)$ below.

Solutions of the Swift–Hohenberg equation have been studied extensively in the pattern formation literature, in the context of both domain-filling and localized patterns. We note that both (1.1) and (1.2) are reversible under $x \mapsto -x$, while only (1.2) is equivariant under the $\mathbb{Z}_2$ symmetry $U \mapsto -U$. Moreover, both systems admit a variational structure.

As was first suggested by Kirchgässner [15] and is now routine, we follow a spatial dynamics approach, analyzing evolution in the spatial variable x. Since we are looking for stationary localized solutions, we consider

$$0 = -(1+\Delta)^2 U - \mu U + g(U,\nu). \tag{1.3}$$

When considering patterns on the line, this may then be recast as the 4-dimensional ODE

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1-\mu & 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ g(u_1,\nu) \end{pmatrix} \tag{1.4}$$

with $u = (u_1,u_2,u_3,u_4)^T = (U, U_x, U_{xx}, U_{xxx})^T \in \mathbb{R}^4$. When considering patterns on the plane which are localized in one direction (x) and periodic in the second (y), we take $x \in \mathbb{R}$ and $y \in S^1$, and arrive at the infinite dimensional ODE

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -(1+\partial_y^2)^2 - \mu & 0 & -2(1+\partial_y^2) & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ g(u_1,\nu) \end{pmatrix} \tag{1.5}$$

with $u = (u_1,u_2,u_3,u_4)^T = (U, U_x, U_{xx}, U_{xxx})^T \in H^3(S^1) \times H^2(S^1) \times H^1(S^1) \times L^2(S^1)$.

Having found a particular localized solution $U_\ell(x)$ or $U_\ell(x,y)$ of (1.1), we are then interested in its stability. This may be addressed by considering the eigenvalue problem

$$\lambda U = -(1+\Delta)^2 U - \mu U + g'(U_\ell,\nu)U. \tag{1.6}$$

Again this may be reformulated as the ordinary differential equation

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1-\mu+g'(U_\ell(x),\nu) & 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} + \lambda \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \tag{1.7}$$

Writing (1.4) as $u_x = f(u, \mu, \nu)$, we note that (1.7) can be written as

$$u_x = f_u(u_\ell, \mu, \nu)u + \lambda B u. \tag{1.8}$$

Similarly, the planar case may be formulated as

$$\begin{aligned} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}_x = & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -(1+\partial_y^2)^2-\mu+g'(U_\ell(x,y),\nu) & 0 & -2(1+\partial_y^2) & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \\ & + \lambda \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}, \end{aligned} \tag{1.9}$$

which is again of the form (1.8). We emphasize that the above discussion is intended to introduce in a concrete way the systems we will study and approach employed, but our results apply more generally to systems satisfying the hypotheses detailed below.

1.2. Outline and overview of results

We begin in §2 by formulating our results for the simplest case—a dynamical system in $\mathbb{R}^4$. Here we may keep in mind a system such as (1.4). We then extend these results in §3 to the more general case of a system in $\mathbb{R}^{2n}$, which contains the essential ingredients for generalization to an infinite dimensional dynamical system addressing patterns in the plane, as in (1.5). In brief, we show that heteroclinic connections (fronts and backs) from a fixed point at the origin to a family of periodic orbits give rise to localized solutions (homoclinic to the origin), which are symmetric if the front and back have the same phase, and asymmetric if they do not. The phase refers to the particular strong stable and unstable fibers of the periodic orbit that intersect the unstable and stable manifolds to the origin at fixed incoming and outgoing sections. Fig. 4 gives a schematic view of our construction, in which a localized solution is constructed from a front with phase φ_1 and a back with phase φ_2. The precise results are given in Theorem 1 and Theorem 2. To illustrate our results more concretely, in §4 we provide numerical computations of planar stripe and spot patterns of the cubic-quintic Swift–Hohenberg equation.

As already mentioned, these existence results are necessary for obtaining rigorous stability results. In §5 we provide context for our stability results and an overview of our approach. Theorem 3 in §6 contains our main stability result, essentially showing that for localized structures,

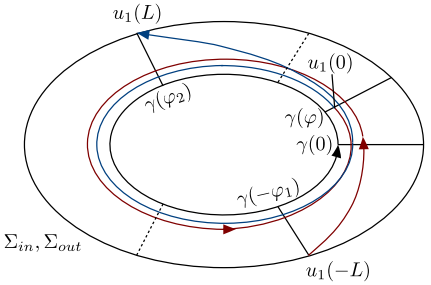


Fig. 4. A schematic view of the analytic gluing construction, with the localized solution $u_1(x)$ shown in red for $x \leq 0$ and in blue for $x \geq 0$, indicating the time periods over which it remains exponentially close to the front and back, respectively. At $x = -L$, the localized solution $u_1(x)$ passes through the section Σ_{in} exponentially close (in L) to the point where the unstable manifold of the origin, $W^u(0, \mu)$, intersects the (center) stable manifold $W^s(\gamma(x, \mu), \mu))$ of the periodic orbit in the section Σ_{in}; this point is in the strong stable fiber $W^{ss}(\gamma(-\varphi_1, \mu), \mu)$. Meanwhile, at $x = L$, the localized solution $u_1(x)$ passes through the section Σ_{out} exponentially close (in L) to the point where the stable manifold of the origin, $W^s(0, \mu)$, intersects the (center) unstable manifold $W^u(\gamma(x, \mu), \mu))$ of the periodic orbit in the section Σ_{out}; this point is in the strong unstable fiber $W^{uu}(\gamma(\varphi_2, \mu), \mu)$. At $x = 0$, the localized solution has center coordinate $\varphi \approx \frac{\varphi_2 - \varphi_1}{2}$. The sections $\Sigma_{in,out}$ are depicted in the same plane for schematic clarity. The dotted lines indicate the strong stable and unstable fibers $W^{ss}(\gamma(-\varphi_2, \mu), \mu)$ and $W^{uu}(\gamma(\varphi_1, \mu), \mu)$, which, by reversibility, intersect $W^u(0, \mu)$ and $W^s(0, \mu)$ in the sections Σ_{in} and Σ_{out}, and give rise to the family of localized solutions $u_2(x; k)$. Symmetric localized solutions are formed from a front and back with the same phase ($\varphi_1 = \varphi_2 \mod 2\pi$), while asymmetric localized solutions are formed by combining a front and back with different phases ($\varphi_1 \neq \varphi_2 \mod 2\pi$). In this illustration, $k = 1$ and $m = 0$.

temporal eigenvalues of the front and back from which they are formed are added with multiplicity in the right half plane. Finally, in §7 we provide numerical illustrations of our results, followed by concluding remarks in §8.

2. The gluing construction in $\mathbb{R}^4$

Consider the system

$$u_x = f(u, \mu), \tag{2.1}$$

with $u \in \mathbb{R}^4$, $\mu \in \mathbb{R}$, and we assume f is smooth (at least C^2 in u and C^1 in μ). We suppose that this is a reversible, conservative system such that the origin is hyperbolic and there exists a family of symmetric periodic orbits depending smoothly on μ. More precisely:

Hypothesis 2.1. There exists a linear map $\mathcal{R} : \mathbb{R}^4 \to \mathbb{R}^4$ with $\mathcal{R}^2 = I$ and dim Fix $\mathcal{R} = 2$ so that $f(\mathcal{R}u, \mu) = -\mathcal{R}f(u, \mu)$ for all (u, μ).

A solution $u(x)$ is called **symmetric** if $u(x) = \mathcal{R}u(-x)$ for all x or, equivalently, $u(0) \in$ Fix $\mathcal{R}$.

Hypothesis 2.2. There exists a smooth function $\mathcal{H} : \mathbb{R}^4 \times \mathbb{R} \to \mathbb{R}$ with $\mathcal{H}(\mathcal{R}u, \mu) = \mathcal{H}(u, \mu)$ and $\langle \nabla_u \mathcal{H}(u, \mu), f(u, \mu) \rangle = 0$ for all (u, μ).

We remark that such a conserved quantity exists whenever the underlying PDE is a gradient system whose energy is given in terms of a Lagrangian; see [19, §3.1] and references therein for

details. In particular, Hypothesis 2.2 holds for the ODE system associated to either form of the Swift–Hohenberg equation.

Hypothesis 2.3. For all μ, $f(0,\mu) = 0$ and $Re\ \sigma f_u(0,\mu) \subset (-\infty, \alpha^s) \cup (\alpha^u, \infty)$ for $\alpha^s < 0 < \alpha^u$.

We normalize the function $\mathcal{H}$ from Hypothesis 2.2 such that $\mathcal{H}(0,\mu) = 0$ for all μ. Moreover, due to Hypothesis 2.1, the spectrum of $f_u(u,\mu)$ is invariant under multiplication by -1, so we may take $\alpha^s = -\alpha^u$ in Hypothesis 2.3.

Hypothesis 2.4. There is a closed interval $J \subset \mathbb{R}$ with nonempty interior such that (2.1) has, for each $\mu \in J$, a periodic orbit $\gamma(x,\mu)$ with minimal period $l(\mu)$ such that:

(i) The family $\gamma(x,\mu)$ depends smoothly on $\mu \in J$.
(ii) $\gamma(x,\mu)$ is symmetric with $\gamma(0,\mu) \in \mathrm{Fix}\,\mathcal{R}$ for all $\mu \in J$.
(iii) For each $\mu \in J$, $\mathcal{H}(\gamma(x,\mu),\mu) = 0$ and $\mathcal{H}_u(\gamma(x,\mu),\mu) \neq 0$ for one and hence all x.
(iv) $\gamma(x,\mu)$ has two positive, nontrivial Floquet multipliers $e^{\pm 2\pi\alpha(\mu)}$ with $\alpha(\mu) > 0$ for all $\mu \in J$.

By Hypotheses 2.1 and 2.2, there will be two Floquet multipliers at one due to translation symmetry and the existence of a conserved quantity (or reversibility). Hypothesis 2.4(iv) requires that the remaining two Floquet multipliers are not unity and, in fact, positive so that the stable and unstable manifolds of the periodic orbits are orientable. We assume without loss of generality that the minimal period $l(\mu) = 2\pi$ for all μ.

Lemma 2.5. *([4]) Assume that Hypotheses 2.1–2.4 are met. Then there exist $\delta > 0$, a smooth reversible change of coordinates near $\gamma(\cdot,\mu)$ and smooth real-valued functions h^c, h^s_j, and h^u_j, $j = 1, 2$, so that (2.1) restricted to the zero energy level set is for all $\mu \in J$ of the form*

$$\begin{aligned} v^c_x &= 1 + h^c(v,\mu)v^s v^u, \\ v^s_x &= -[\alpha(\mu) + h^s_1(v,\mu)v^s + h^s_2(v,\mu)v^u]v^s, \\ v^u_x &= [\alpha(\mu) + h^u_1(v,\mu)v^s + h^u_2(v,\mu)v^u]v^u, \end{aligned} \tag{2.2}$$

where $v = (v^c, v^s, v^u) \in \mathcal{V} := S^1 \times I \times I$ and $I = [-\delta, \delta]$. $\mathcal{R}$ acts on v via

$$\mathcal{R}(v^c, v^s, v^u) = (-v^c, v^s, v^u). \tag{2.3}$$

In this coordinate system, we define the sections

$$\Sigma_{in} := S^1 \times \{v^s = \delta\} \times I, \quad \Sigma_{out} := S^1 \times I \times \{v^u = \delta\}. \tag{2.4}$$

We also define the set

$$\Gamma := \{(\varphi,\mu) \in S^1 \times J : W^s(0,\mu) \cap W^{uu}(\gamma(\varphi,\mu),\mu) \cap \Sigma_{out} \neq \emptyset\}. \tag{2.5}$$

We now assume Γ is the graph of a smooth function, and that locally near each heteroclinic orbit, the intersection of the stable manifold $W^s(0,\mu)$ with the section Σ_{out} is also described by a smooth function.

Hypothesis 2.6. The set Γ is the graph of a smooth function $z : S^1 \to \mathring{J}$. Furthermore, there exist an open neighborhood U_Γ of Γ in $S^1 \times J$, positive constants $\epsilon, b > 0$, and a smooth function $g : U_\Gamma \to I$ so that

$$\begin{aligned} &\{(\varphi, v^s, \delta) \in W^s(0,\mu) \cap \Sigma_{out} : |v^s| < \epsilon, (\varphi,\mu) \in U_\Gamma\} \\ &\quad = \{(\varphi, v^s, \delta) = (\varphi, g(\varphi,\mu), \delta) : (\varphi,\mu) \in U_\Gamma\} \end{aligned} \tag{2.6}$$

and $|g_\mu(\varphi,\mu)| \geq b > 0$ for all $(\varphi,\mu) \in U_\Gamma$.

Lemma 2.7. *([4]) There exist positive constants L_0 and η so that the following is true for all $L > L_0$ and $\varphi \in S^1$: there is a unique solution $v(x)$ of (2.2), defined for $x \in [-L, L]$ such that*

$$v(-L) \in \Sigma_{in}, \;\; v(L) \in \Sigma_{out}, \;\; v^c(0) = \varphi, \;\; v(x) \in \mathcal{V} \;\; \forall x \in [-L, L].$$

Furthermore, we have

$$\begin{aligned} v(-L) &= \left(\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, \delta e^{-2\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right)\right) \\ v(0) &= \left(\varphi, \delta e^{-\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right), \delta e^{-\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right)\right) \\ v(L) &= \left(\varphi + L + \mathrm{O}(e^{-\eta L}), \delta e^{-2\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right), \delta\right). \end{aligned} \tag{2.7}$$

The solution $v(x)$ is smooth in (L, φ, μ) and the error estimates in (2.7) can be differentiated.

We now show that fronts and backs, i.e., solutions that approach the homogeneous rest state in backward (respectively forward) time and the periodic orbit γ in forward (respectively backward) time, can be "glued" together to form both symmetric and asymmetric localized solutions. Moreover, these solutions depend continuously on the phases φ_1 and φ_2 of the associated fronts and backs. We start by introducing and parameterizing the front and back solutions in a convenient fashion.

For each $\varphi \in S^1$, we define $u_f(x;\varphi)$ to be the solution such that

$$u_f(0;\varphi) \in \Sigma_{in} \cap W^u(0, z(\varphi)) \cap W^{ss}(\gamma(-\varphi, z(\varphi)), z(\varphi)). \tag{2.8}$$

Further defining $u_b(x;\varphi) := \mathcal{R}u_f(-x;\varphi)$, we have

$$u_b(0;\varphi) \in \Sigma_{out} \cap W^s(0, z(\varphi)) \cap W^{uu}(\gamma(\varphi, z(\varphi)), z(\varphi)). \tag{2.9}$$

We note that these definitions imply

$$|u_f(x;\varphi) - \gamma(x - \varphi)| \leq Ce^{-\eta x}, \quad x \geq 0 \tag{2.10}$$

$$|u_b(x;\varphi) - \gamma(x + \varphi)| \leq Ce^{\eta x}, \quad x \leq 0 \tag{2.11}$$

with positive constants $C, \eta > 0$ that do not depend on φ.

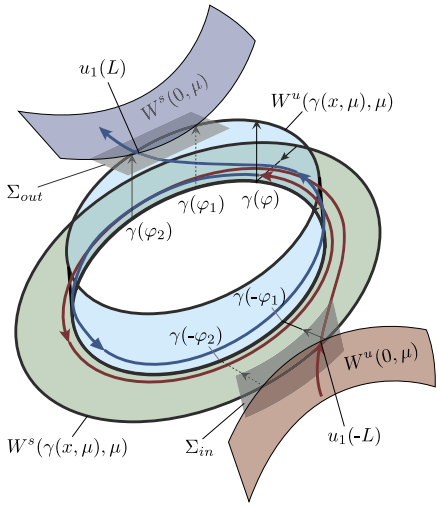


Fig. 5. A three-dimensional schematic view of the analytic gluing construction, including the manifolds and sections of interest. Of course, this schematic is only valid locally as the three dimensional manifold is immersed in four dimensional space. The localized solution $u_1(x)$ is shown in red for $x \leq 0$ and in blue for $x \geq 0$, indicating the time periods over which it remains exponentially close to the front and back, respectively. $u_1(-L)$ passes through the section Σ_{in} exponentially close (in L) to the point where the unstable manifold of the origin, $W^u(0,\mu)$, intersects the (center) stable manifold $W^s(\gamma(x,\mu),\mu))$ of the periodic orbit in the section Σ_{in}; this point is in the strong stable fiber $W^{ss}(\gamma(-\varphi_1,\mu),\mu)$. Meanwhile, $u_1(L)$ passes through the section Σ_{out} exponentially close (in L) to the point where the stable manifold of the origin, $W^s(0,\mu)$, intersects the (center) unstable manifold $W^u(\gamma(x,\mu),\mu))$ of the periodic orbit in the section Σ_{out}; this point is in the strong unstable fiber $W^{uu}(\gamma(\varphi_2,\mu),\mu)$. At $x=0$ (indicated by the change from red to blue coloring), $u_1(0)$ has center coordinate $\varphi \approx \frac{\varphi_2-\varphi_1}{2}$ (see (2.15)). Dotted lines indicate the strong stable and unstable fibers $W^{ss}(\gamma(-\varphi_2,\mu),\mu)$ and $W^{uu}(\gamma(\varphi_1,\mu),\mu)$, which, by reversibility, intersect $W^u(0,\mu)$ and $W^s(0,\mu)$ in the sections Σ_{in} and Σ_{out}, as seen in the figure, and give rise to the family of localized solutions $u_2(x;k)$. Symmetric localized solutions are formed from a front and back with the same phase ($\varphi_1 = \varphi_2 \mod 2\pi$), while asymmetric localized solutions are formed by combining a front and back with different phases ($\varphi_1 \neq \varphi_2 \mod 2\pi$). In this illustration, $k=1$ and $m=0$.

In the coordinate system introduced in Lemma 2.5 near the periodic orbit γ, we record that

$$u_f(0;\varphi) = (-\varphi, \delta, 0), \tag{2.12}$$

$$u_b(0;\varphi) = (\varphi, 0, \delta). \tag{2.13}$$

In Fig. 5 we show a schematic illustration of the localized solution whose existence and uniqueness we prove in the following:

Theorem 1 *(Gluing fronts and backs to produce localized solutions). Fix $\epsilon > 0$, then there exist $\eta, K, C > 0$ such that for each $\varphi_1^*, \varphi_2^* \in S^1$ with $\mu = z(\varphi_1^*) = z(\varphi_2^*)$ and $|z'(\varphi_1^*)|, |z'(\varphi_2^*)| \geq \epsilon$, and for each $k \in \mathbb{N}, k \geq K$, and $m \in \{0,1\}$, we can define $\varphi_1 = \varphi_1^*$, $\varphi_2 = \varphi_2^* + 2m\pi$, and*

$$L = L(\varphi_1, \varphi_2, k) := \frac{\varphi_1 + \varphi_2}{2} + 2k\pi + \mathrm{O}(e^{-\eta 2k\pi}), \tag{2.14}$$

$$\varphi = \varphi(\varphi_1, \varphi_2, k) := \frac{\varphi_2 - \varphi_1}{2} + \mathrm{O}(e^{-\eta 2k\pi}) \tag{2.15}$$

such that there exists a unique solution $u_1(x) = u_1(x;\varphi_1,\varphi_2,k)$ of (2.1), depending smoothly on (φ_1,φ_2) for each k, with $\lim_{|x|\to\infty}|u_1(x)| = 0$, $u_1(x) \in \mathcal{V}$ for $x \in [-L,L]$, and in the coordinate system introduced in Lemma 2.5 near the periodic orbit γ we have

$$\begin{aligned} u_1(-L) &= \left(\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, \delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right)\right) \\ u_1(0) &= \left(\varphi, \delta e^{-\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right), \delta e^{-\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right)\right) \\ u_1(L) &= \left(\varphi + L + \mathrm{O}(e^{-\eta L}), \delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right), \delta\right), \end{aligned} \tag{2.16}$$

where the first coordinate is understood to be taken modulo 2π. Moreover,

$$\left|u_1(x) - u_f(x+L;\varphi_1)\right| \le Ce^{\eta x}, \qquad x \le -L \tag{2.17}$$

$$\left|u_1(x) - u_f(x+L;\varphi_1)\right| \le Ce^{-\eta L}, \quad x \in [-L,0] \tag{2.18}$$

$$|u_1(x) - u_b(x-L;\varphi_2)| \le Ce^{-\eta L}, \quad x \in [0,L] \tag{2.19}$$

$$|u_1(x) - u_b(x-L;\varphi_2)| \le Ce^{-\eta x}, \quad x \ge L \tag{2.20}$$

Similarly, there exists a unique solution $u_2(x) = u_2(x;\varphi_1,\varphi_2,k)$ of (2.1), depending smoothly on (φ_1,φ_2) for each k, and with L and φ as in (2.14) and (2.15), such that $\lim_{|x|\to\infty}|u_2(x)| = 0$, $u_2(x) \in \mathcal{V}$ for $x \in [-L,L]$, and

$$\begin{aligned} u_2(-L) &= \left(-\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, \delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right)\right) \\ u_2(0) &= \left(-\varphi, \delta e^{-\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right), \delta e^{-\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right)\right) \\ u_2(L) &= \left(-\varphi + L + \mathrm{O}(e^{-\eta L}), \delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right), \delta\right), \end{aligned} \tag{2.21}$$

where again the first coordinate is understood to be taken modulo 2π. Moreover,

$$\left|u_2(x) - u_f(x+L;\varphi_2)\right| \le Ce^{\eta x}, \qquad x \le -L \tag{2.22}$$

$$\left|u_2(x) - u_f(x+L;\varphi_2)\right| \le Ce^{-\eta L}, \quad x \in [-L,0] \tag{2.23}$$

$$|u_2(x) - u_b(x-L;\varphi_1)| \le Ce^{-\eta L}, \quad x \in [0,L] \tag{2.24}$$

$$|u_2(x) - u_b(x-L;\varphi_1)| \le Ce^{-\eta x}, \quad x \ge L \tag{2.25}$$

Remark 2.8. By setting $\varphi_1^* = \varphi_2^*$ and $m = 0$, we recover the branch of symmetric solutions with center coordinate $\varphi = 0$ at $x = 0$. By setting $\varphi_1^* = \varphi_2^*$ and $m = 1$, we recover the second branch of symmetric solutions with center coordinate $\varphi = \pi$ at $x = 0$. In each case the families of solutions $u_1(x)$ and $u_2(x)$ are the same.

Remark 2.9. It is evident from the proof of Theorem 1 that we have a unique gluing solution for any φ_1^*, φ_2^* with $z(\varphi_1^*) = z(\varphi_2^*)$ and $z'(\varphi_1^*), z'(\varphi_2^*) \neq 0$. We fix $\epsilon > 0$ only in order to have uniform constants η, K, C. Thus we have the following corollary:

Corollary 2.10. *For any φ_1^*, φ_2^* with $\mu = z(\varphi_1^*) = z(\varphi_2^*)$ and $|z'(\varphi_1^*)|, |z'(\varphi_2^*)| \neq 0$ there exist $\eta, K, C > 0$ such that for each $k \geq K$ and $m \in \{0, 1\}$, there exists a unique solution $u_1(x)$ satisfying the conditions in Theorem 1. The solution $u_2(x)$ is distinct from $u_1(x)$ provided that $\varphi_1^* \neq \varphi_2^*$.*

Proof of Theorem 1. We first prove all statements for $u_1(x)$. In order for a solution $u_1(x)$ to satisfy $\lim_{|x|\to\infty} |u_1(x)| = 0$ and $u_1(\pm L) \in \mathcal{V}$, we must have

$$u_1(-L) \in \Sigma_{in} \cap W^u(0, z(\varphi_1)), \quad u_1(L) \in \Sigma_{out} \cap W^s(0, z(\varphi_2)). \tag{2.26}$$

[Recall $z(\varphi_1) = z(\varphi_2)$.] From Hypothesis 2.6 we have that $(\varphi, v^s, \delta) \in W^s(0, \mu) \cap \Sigma_{out}$ if and only if $v^s = g(\varphi, \mu)$. Then using (2.3), $(\varphi, \delta, v^u) \in W^u(0, \mu) \cap \Sigma_{in}$ if and only if $\mathcal{R}(\varphi, \delta, v^u) = (-\varphi, v^u, \delta) \in W^s(0, \mu) \cap \Sigma_{out}$ if and only if $v^u = g(-\varphi, \mu)$.

Thus (2.26) is equivalent to

$$u_1^u(-L) = g(-u_1^c(-L), z(\varphi_1)) \tag{2.27}$$

$$u_1^s(L) = g(u_1^c(L), z(\varphi_2)). \tag{2.28}$$

We now let

$$L = L(\varphi_1, \varphi_2, k, \varepsilon_1) := \frac{\varphi_1 + \varphi_2}{2} + 2k\pi + \varepsilon_1, \tag{2.29}$$

$$\varphi = \varphi(\varphi_1, \varphi_2, k, \varepsilon_2) := \frac{\varphi_2 - \varphi_1}{2} + \varepsilon_2. \tag{2.30}$$

We will show that ε_1 and ε_2 can be chosen such that (2.26) holds and both are $\mathrm{O}(e^{-\eta 2k\pi})$. We note that $L + \varphi = \varphi_2 + \varepsilon_1 + \varepsilon_2 \bmod 2k\pi$ and $L - \varphi = \varphi_1 + \varepsilon_1 - \varepsilon_2 \bmod 2k\pi$. So by Lemma 2.7, the unique solution associated to (L, φ) as defined in (2.29)–(2.30) has

$$u_1(-L) = \left(-\varphi_1 - \varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L}), \delta, \delta e^{-2\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right)\right) \tag{2.31}$$

$$u_1(0) = \left(\varphi, \delta e^{-\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right), \delta e^{-\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right)\right) \tag{2.32}$$

$$u_1(L) = \left(\varphi_2 + \varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L}), \delta e^{-2\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right), \delta\right). \tag{2.33}$$

Thus $u_1^c(-L) - u_f^c(0; \varphi_1) = -\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L})$. To solve (2.27), we substitute from (2.31) and expand $g(-u_1^c(-L), z(\varphi_1))$ about $(-u_f^c(0; \varphi_1), z(\varphi_1))$, arriving at

$$\begin{aligned}\delta e^{-2\alpha(\mu)L}\left(1 + \mathrm{O}(e^{-\eta L})\right) = {} & g(-u_f^c(0; \varphi_1), z(\varphi_1)) \\ & + g_\varphi(-u_f^c(0; \varphi_1), z(\varphi_1))(-\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L})) \\ & + \mathrm{O}((-\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L}))^2).\end{aligned} \tag{2.34}$$

By Hypothesis 2.6,

$$g(\varphi, z(\varphi)) \equiv 0 \tag{2.35}$$

so $g(-u_f^c(0;\varphi_1), z(\varphi_1)) = g(\varphi_1, z(\varphi_1)) = 0$. Thus

$$\varepsilon_1 = -\frac{\delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right) + \mathrm{O}((-\varepsilon_1+\varepsilon_2+\mathrm{O}(e^{-\eta L}))^2)}{g_\varphi(\varphi_1, z(\varphi_1))} + \varepsilon_2 + \mathrm{O}(e^{-\eta L}). \tag{2.36}$$

Now returning to (2.28), we substitute from (2.33), note that $u_1^c(L) - u_b^c(0;\varphi_2) = \varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L})$ and expand $g(u_1^c(L), z(\varphi_2))$ about $(u_b^c(0;\varphi_2), z(\varphi_2))$ to get

$$\begin{aligned}\delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right) &= g(u_f^c(0;\varphi_2), z(\varphi_2)) + g_\varphi(u_f^c(0;\varphi_2), z(\varphi_2))(-\varepsilon_1-\varepsilon_2+\mathrm{O}(e^{-\eta L})) \\ &\quad + \mathrm{O}((\varepsilon_1+\varepsilon_2+\mathrm{O}(e^{-\eta L}))^2).\end{aligned} \tag{2.37}$$

Then using $g(u_f^c(0;\varphi_2), z(\varphi_2)) = 0$ as before,

$$\varepsilon_1 + \varepsilon_2 = \frac{\delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right) + \mathrm{O}((\varepsilon_1+\varepsilon_2+\mathrm{O}(e^{-\eta L}))^2)}{g_\varphi(\varphi_2, z(\varphi_2))} + \mathrm{O}(e^{-\eta L}) \tag{2.38}$$

so that combining this with (2.36), we have

$$\begin{aligned}\varepsilon_2 = \frac{1}{2}\Bigg[&\frac{\delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right) + \mathrm{O}((\varepsilon_1+\varepsilon_2+\mathrm{O}(e^{-\eta L}))^2)}{g_\varphi(\varphi_2, z(\varphi_2))} \\ &+ \frac{\delta e^{-2\alpha(\mu)L}\left(1+\mathrm{O}(e^{-\eta L})\right) + \mathrm{O}((-\varepsilon_1+\varepsilon_2+\mathrm{O}(e^{-\eta L}))^2)}{g_\varphi(\varphi_1, z(\varphi_1))}\Bigg] + \mathrm{O}(e^{-\eta L}).\end{aligned} \tag{2.39}$$

Now by (2.35), $g_\varphi(\varphi_1, z(\varphi_1)) = -g_\mu(\varphi_1, z(\varphi_1))z'(\varphi_1)$ and similarly for φ_2, so that by our original assumption combined with Hypothesis 2.6, $|g_\varphi(\varphi_1, z(\varphi_1))|$, $|g_\varphi(\varphi_1, z(\varphi_1))| > b\epsilon$. Thus we have $\varepsilon_1, \varepsilon_2$ in (2.29)–(2.30) so that (2.26) holds, and both $\varepsilon_1, \varepsilon_2$ are $\mathrm{O}(e^{-\eta L})$.

Now since we have a solution $u_1(x)$ such that $u_1(-L) = (\varphi_1 + \mathrm{O}(e^{-\eta L}), \delta, \delta e^{-2\alpha(\mu)L}(1 + \mathrm{O}(e^{-\eta L})))$, it follows that $|u_1(-L) - u_f(0;\varphi_1)| = \mathrm{O}(e^{-\eta L})$. Similarly, $|u_1(L) - u_b(0;\varphi_2)| = \mathrm{O}(e^{-\eta L})$. Now since both $u_1(-L) \in W^u(0,\mu)$ and $u_f(-\varphi) \in W^u(0,\mu)$, the exponential approach in backward time expressed in (2.17) follows from the stable manifold theorem. Similarly, $u_1(L) \in W^s(0,\mu)$ and $u_b(\varphi_2) \in W^s(0,\mu)$ implies (2.20). On the other hand, from the expression (2.2) we have that $|u_1(x) - u_f(x+L;\varphi_1)| = O(e^{-\eta L})$ on $-L \le x \le 0$, so that for K sufficiently large, there exists a constant C such that for all $L(k)$ with $k \ge K$, (2.18) is satisfied. The relation (2.19) is shown analogously.

The claims for $u_2(x)$ follow upon setting $u_2(x) = \mathcal{R}u_1(-x)$. In particular, the expressions in (2.21) follow from those in (2.16) by using, for example, $u_2(-L) = \mathcal{R}u_1(L)$ along with the relation $\mathcal{R}(v^c, v^s, v^u) = (-v^c, v^u, v^s)$. Furthermore,

$$|u_1(x) - u_f(x+L;\varphi_1)| \le Ce^{\eta x}, \quad x \le -L$$

implies

$$|u_1(-y) - u_f(-y+L;\varphi_1)| \le Ce^{-\eta y}, \quad y \ge L$$

$$|\mathcal{R}u_1(-y) - \mathcal{R}u_f(-y+L;\varphi_1)| \le Ce^{-\eta y}, \quad y \ge L$$
$$|u_2(y) - u_b(y-L;\varphi_1)| \le Ce^{-\eta y}, \quad y \ge L$$

upon employing the coordinate transform $y = -x$ and applying $\mathcal{R}$, so that (2.17) implies (2.25) and the other relations follow similarly. □

While the proof of Theorem 1 is complete as presented, in Appendix A we explicitly construct solutions near the periodic orbits for $[-L, L]$ rather than invoking Lemma 2.7. We believe this construction clarifies the result and the pieces needed for extension to $\mathbb{R}^{2n}$ and appropriate Banach spaces.

3. The gluing construction in $\mathbb{R}^{2n}$

We now consider the system

$$u_x = f(u, \mu), \tag{3.1}$$

with $u \in \mathbb{R}^{2n}$ and $\mu \in \mathbb{R}$, and again we assume f is a smooth function. The first three hypotheses are completely analogous to those for the system in $\mathbb{R}^4$, and we do not repeat them here. The fourth hypothesis is modified as follows:

Hypothesis 3.1. There is a closed interval $J \subset \mathbb{R}$ with nonempty interior such that (2.1) has, for each $\mu \in J$, a periodic orbit $\gamma(x, \mu)$ with minimal period $l(\mu)$ such that:

(i) The family $\gamma(x, \mu)$ depends smoothly on $\mu \in J$.
(ii) $\gamma(x, \mu)$ is symmetric with $\gamma(0, \mu) \in \operatorname{Fix} \mathcal{R}$ for all $\mu \in J$.
(iii) For each $\mu \in J$, $\mathcal{H}(\gamma(x, \mu), \mu) = 0$ and $\mathcal{H}_u(\gamma(x, \mu), \mu) \neq 0$ for one and hence all x.
(iv) $\gamma(x, \mu)$ has two simple Floquet multipliers, $e^{\pm 2\pi\alpha(\mu)}$, with $\alpha(\mu) > 0$ for all $\mu \in J$, and there is an $\eta > 0$ such that the remaining Floquet multipliers have modulus less than $e^{-2\pi(\alpha(\mu)+\eta)}$ or greater than $e^{2\pi(\alpha(\mu)+\eta)}$ uniformly in $\mu \in J$.

We again assume without loss of generality that the minimal period $l(\mu) = 2\pi$ for all μ. In this case, for $\delta > 0$ sufficiently small we can write the space $\mathcal{V}$ in the zero energy level set near $\gamma(x, \mu)$ as

$$\mathcal{V} := S^1 \times I \times B \times I \times B, \quad I = [-\delta, \delta], \quad B = B_\delta(0) \in \mathbb{R}^{n-2} \tag{3.2}$$

with the corresponding coordinate system

$$v = (v^c, v^s, v^{ss}, v^u, v^{uu}), \tag{3.3}$$

so that within $\mathcal{V}$, $W^{ss}(\gamma(\varphi, \mu), \mu)) = (\varphi, v^s, v^{ss}, 0, 0)$ and $W^{uu}(\gamma(\varphi, \mu), \mu) = (\varphi, 0, 0, v^u, v^{uu})$.

We then define the sections

$$\Sigma_{in} := S^1 \times \{v^s = \delta\} \times B \times I \times B, \quad \Sigma_{out} := S^1 \times I \times B \times \{v^u = \delta\} \times B. \tag{3.4}$$

As before we define the set

$$\Gamma := \{(\varphi, \mu) \in S^1 \times J : W^s(0, \mu) \cap W^{uu}(\gamma(\varphi, \mu), \mu) \cap \Sigma_{out} \neq \emptyset\}. \tag{3.5}$$

Our final hypothesis is then similar to that for $\mathbb{R}^4$, and additionally enforces that the stable manifold $W^s(0, \mu)$ approaches the unstable manifold $W^u(\gamma(x, \mu), \mu)$ in backward time along the most weakly unstable direction of the flow near the periodic orbit.

Hypothesis 3.2. The set Γ is the graph of a smooth function $z : S^1 \to \mathring{J}$. Furthermore, there exist an open neighborhood U_Γ of Γ in $S^1 \times J$, positive constants $\epsilon, b > 0$, and smooth functions $g : U_\Gamma \times B \to I$ and $h : U_\Gamma \times B \to B$ so that

$$\begin{aligned} &\{(\varphi, v^s, v^{ss}, \delta, v^{uu}) \in W^s(0, \mu) \cap \Sigma_{out} : |v^s| < \epsilon, (\varphi, \mu) \in U_\Gamma, v^{ss} \in B\} \\ &= \{(\varphi, v^s, v^{ss}, \delta, v^{uu}) = (\varphi, g(\varphi, \mu, v^{ss}), v^{ss}, \delta, h(\varphi, \mu, v^{ss})) : (\varphi, \mu) \in U_\Gamma, v^{ss} \in B\} \end{aligned} \tag{3.6}$$

and $|g_\mu(\varphi, \mu, v^{ss})| \geq b > 0$ for all $(\varphi, \mu, v^{ss}) \in U_\Gamma \times B$.

Under these conditions, we have the following analogue of Lemma 2.7:

Lemma 3.3. *(See Appendix A and Lemma 6.2 in [4]) There exist positive constants L_0 and η so that the following is true for all $L > L_0$ and $\varphi \in S^1$: for each $v^{ss}, v^{uu} \in B$ there is a unique solution $v(x)$ of (3.1), defined for $x \in [-L, L]$ such that*

$$v(-L) \in \Sigma_{in}, \quad v(L) \in \Sigma_{out}, \quad v^c(0) = \varphi, \quad v(x) \in \mathcal{V} \ \forall x \in [-L, L].$$

Furthermore, we have

$$\begin{aligned} v(-L) &= \left(\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, \mathrm{O}(e^{-\eta L}), v^{ss} + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\right) \\ v(0) &= \left(\varphi, \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\right) \\ v(L) &= \left(\varphi + L + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \delta, v^{uu} + \mathrm{O}(e^{-\eta L})\right). \end{aligned} \tag{3.7}$$

The solution $v(x)$ is smooth in $(L, \varphi, \mu, v^{ss}, v^{uu})$ and the error estimates in (3.7) can be differentiated.

We now introduce the front and back solutions. As before, we define $u_f(x; \varphi)$ so that

$$u_f(0; \varphi) \in \Sigma_{in} \cap W^u(0, z(\varphi)) \cap W^{ss}(\gamma(-\varphi, z(\varphi)), z(\varphi)), \tag{3.8}$$

and with $u_b(x; \varphi) := \mathcal{R} u_f(-x; \varphi)$, we have

$$u_b(0; \varphi) \in \Sigma_{out} \cap W^s(0, z(\varphi)) \cap W^{uu}(\gamma(\varphi, z(\varphi)), z(\varphi)). \tag{3.9}$$

By Hypothesis 3.2 we then have

$$u_f(0;\varphi) = (-\varphi, \delta, h(\varphi, z(\varphi), 0), 0, 0) \tag{3.10}$$

$$u_b(0;\varphi) = (\varphi, 0, 0, \delta, h(\varphi, z(\varphi), 0)). \tag{3.11}$$

Under these conditions we have the following analogue of Theorem 1:

Theorem 2. *Fix $\epsilon > 0$, then there exist $\eta, K, C > 0$ such that for each $\varphi_1^*, \varphi_2^* \in S^1$ with $\mu = z(\varphi_1^*) = z(\varphi_2^*)$ and $|z'(\varphi_1^*)|, |z'(\varphi_2^*)| \geq \epsilon$, and for each $k \in \mathbb{N}, k \geq K$, and $m \in \{0, 1\}$, we can define $\varphi_1 = \varphi_1^*$, $\varphi_2 = \varphi_2^* + 2m\pi$, and*

$$L = L(\varphi_1, \varphi_2, k) := \frac{\varphi_1 + \varphi_2}{2} + 2k\pi + \mathrm{O}(e^{-\eta 2k\pi}), \tag{3.12}$$

$$\varphi = \varphi(\varphi_1, \varphi_2, k) := \frac{\varphi_2 - \varphi_1}{2} + \mathrm{O}(e^{-\eta 2k\pi}) \tag{3.13}$$

such that there exists a unique solution $u_1(x) = u_1(x; \varphi_1, \varphi_2, k)$ of (3.1), depending smoothly on (φ_1, φ_2) for each k, with $\lim_{|x|\to\infty} |u_1(x)| = 0$, $u_1(x) \in \mathcal{V}$ for $x \in [-L, L]$, and in the coordinate system introduced in (3.3) near the periodic orbit γ we have

$$\begin{aligned}
u_1(-L) &= \Big(\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, h(\varphi_1, 0, z(\varphi_1)) + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\Big) \\
u_1(0) &= \Big(\varphi, \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\Big) \\
u_1(L) &= \Big(\varphi + L + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \delta, h(\varphi_2, 0, z(\varphi_2)) + \mathrm{O}(e^{-\eta L})\Big),
\end{aligned} \tag{3.14}$$

where the first coordinate is understood to be taken modulo 2π. Moreover,

$$\big|u_1(x) - u_f(x + L; \varphi_1)\big| \leq Ce^{\eta x}, \quad x \leq -L \tag{3.15}$$

$$\big|u_1(x) - u_f(x + L; \varphi_1)\big| \leq Ce^{-\eta L}, \quad -L \leq x \leq 0 \tag{3.16}$$

$$|u_1(x) - u_b(x - L; \varphi_2)| \leq Ce^{-\eta L}, \quad 0 \leq x \leq L \tag{3.17}$$

$$|u_1(x) - u_b(x - L; \varphi_2)| \leq Ce^{-\eta x}, \quad x \geq L \tag{3.18}$$

Similarly, there exists a unique solution $u_2(x) = u_2(x; \varphi_1, \varphi_2, k)$ of (2.1), depending smoothly on (φ_1, φ_2) for each k, and with L and φ as in (3.12) and (3.13), such that $\lim_{|x|\to\infty} |u_2(x)| = 0$, $u_2(x) \in \mathcal{V}$ for $x \in [-L, L]$, and

$$\begin{aligned}
u_2(-L) &= \Big(-\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, h(\varphi_2, 0, z(\varphi_2)) + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\Big) \\
u_2(0) &= \Big(-\varphi, \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\Big) \\
u_2(L) &= \Big(-\varphi + L + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \delta, h(\varphi_1, 0, z(\varphi_1)) + \mathrm{O}(e^{-\eta L})\Big),
\end{aligned} \tag{3.19}$$

where again the first coordinate is understood to be taken modulo 2π. Moreover,

$$\left|u_2(x) - u_f(x+L;\varphi_2)\right| \le Ce^{\eta x}, \quad x \le -L \tag{3.20}$$

$$\left|u_2(x) - u_f(x+L;\varphi_2)\right| \le Ce^{-\eta L}, \quad -L \le x \le 0 \tag{3.21}$$

$$|u_2(x) - u_b(x-L;\varphi_1)| \le Ce^{-\eta L}, \quad 0 \le x \le L \tag{3.22}$$

$$|u_2(x) - u_b(x-L;\varphi_1)| \le Ce^{-\eta x}, \quad x \ge L \tag{3.23}$$

Proof. As before, we wish to show that for L and φ as in (3.12)–(3.13) we can satisfy

$$u_1(-L) \in \Sigma_{in} \cap W^u(0, z(\varphi_1)), \quad u_1(L) \in \Sigma_{out} \cap W^s(0, z(\varphi_2)). \tag{3.24}$$

We note that $(\varphi, \delta, v^{ss}, v^u, v^{uu}) \in \Sigma_{in} \cap W^u(0,\mu)$ if and only if

$$\mathcal{R}(\varphi, \delta, v^{ss}, v^u, v^{uu}) = (-\varphi, v^u, v^{uu}, v^s, v^{ss}) \in \Sigma_{out} \cap W^s(0,\mu). \tag{3.25}$$

So by Hypothesis 3.2, the conditions in (3.24) are equivalent to the $2n-2$ conditions

$$u_1^u(-L) = g(-u_1^c(-L), z(\varphi_1), u_1^{uu}(-L)) \tag{3.26}$$

$$u_1^{ss}(-L) = h(-u_1^c(-L), z(\varphi_1), u_1^{uu}(-L)) \tag{3.27}$$

$$u_1^s(L) = g(u_1^c(L), z(\varphi_2), u_1^{ss}(L)) \tag{3.28}$$

$$u_1^{uu}(L) = h(u_1^c(L), z(\varphi_2), u_1^{ss}(L)) \tag{3.29}$$

Define

$$L = L(\varphi_1, \varphi_2, k, \varepsilon_1) := \frac{\varphi_1 + \varphi_2}{2} + 2k\pi + \varepsilon_1 \tag{3.30}$$

$$\varphi = \varphi(\varphi_1, \varphi_2, \varepsilon_2) := \frac{\varphi_2 - \varphi_1}{2} + \varepsilon_2 \tag{3.31}$$

$$w^{ss} = w^{ss}(\varphi_1, \varepsilon_3) := h(\varphi_1, z(\varphi_1), 0) + \varepsilon_3 \tag{3.32}$$

$$w^{uu} = w^{uu}(\varphi_2, \varepsilon_4) := h(\varphi_2, z(\varphi_2), 0) + \varepsilon_4 \tag{3.33}$$

with $\varepsilon_1, \varepsilon_2 \in \mathbb{R}$ and $\varepsilon_3, \varepsilon_4 \in \mathbb{R}^{n-2}$. Let $u_1(x) = u_1(x; L, \varphi, w^{ss}, w^{uu})$ be the unique solution such that $u_1(x) \in \mathcal{V}$ for $x \in [-L, L]$ and

$$\begin{aligned} u_1(-L) &= \left(\varphi - L + \mathrm{O}(e^{-\eta L}), \delta, w^{ss} + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\right) \\ u_1(0) &= \left(\varphi, \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L})\right) \\ u_1(L) &= \left(\varphi + L + \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \mathrm{O}(e^{-\eta L}), \delta, w^{uu} + \mathrm{O}(e^{-\eta L})\right). \end{aligned} \tag{3.34}$$

Then substituting from (3.34) into (3.26), Taylor expanding about

$$(-u_f^c(0;\varphi_1, z(\varphi_1), u_f^{uu}(0;\varphi_1)) = (\varphi_1, z(\varphi_1), 0),$$

and using (3.30)–(3.31), we have

$$\begin{aligned}\mathrm{O}(e^{-\eta L}) = g(\varphi_1, z(\varphi_1), 0) + g_{v^c}(\varphi_1, z(\varphi_1), 0)(\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L})) \\ + g_{v^{ss}}(\varphi_1, z(\varphi_1), 0)\mathrm{O}(e^{-\eta L}) + \mathrm{O}((\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L}))^2) + \mathrm{O}(e^{-2\eta L}). \quad (3.35)\end{aligned}$$

By Hypothesis 3.2, $g(\varphi_1, z(\varphi_1), 0) = 0$, so

$$\varepsilon_1 = \frac{\mathrm{O}(e^{-\eta L})(1 + g_{v^{ss}}(\varphi_1, z(\varphi_1), 0)) + \mathrm{O}((\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L}))^2)}{g_{v^c}(\varphi_1, z(\varphi_1), 0)} + \varepsilon_2 + \mathrm{O}(e^{-\eta L}). \quad (3.36)$$

Similarly, substituting from (3.34) into (3.28), Taylor expanding about

$$(u_b^c(0; \varphi_2), z(\varphi_2), u_b^{ss}(0; \varphi_2)) = (\varphi_2, z(\varphi_2), 0),$$

and using (3.30)–(3.31), we have

$$\begin{aligned}\mathrm{O}(e^{-\eta L}) = g(\varphi_2, z(\varphi_2), 0) + g_{v^c}(\varphi_2, z(\varphi_2), 0)(\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L})) \\ + g_{v^{ss}}(\varphi_2, z(\varphi_2), 0)\mathrm{O}(e^{-\eta L}) + \mathrm{O}((\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L}))^2) + \mathrm{O}(e^{-2\eta L}) \quad (3.37)\end{aligned}$$

so that

$$\begin{aligned}\varepsilon_2 = \frac{\mathrm{O}(e^{-\eta L})(1 - g_{v^{ss}}(\varphi_2, z(\varphi_2), 0)) + \mathrm{O}((\varepsilon_1 + \varepsilon_2 + \mathrm{O}(e^{-\eta L}))^2)}{g_{v^c}(\varphi_2, z(\varphi_2), 0)} \\ - \frac{\mathrm{O}(e^{-\eta L})(1 + g_{v^{ss}}(\varphi_1, z(\varphi_1), 0)) + \mathrm{O}((\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L}))^2)}{g_{v^c}(\varphi_1, z(\varphi_1), 0)} + \mathrm{O}(e^{-\eta L}). \quad (3.38)\end{aligned}$$

Then since $g(\varphi, z(\varphi), 0) \equiv 0$ for all φ and $|g_\mu(\varphi, z(\varphi), 0)| > 0$ by Hypothesis 3.2, and $|z'(\varphi_i)| > 0$ for $i = 1, 2$ by assumption, both ε_1 and ε_2 are $\mathrm{O}(e^{-\eta L})$. Turning now to ε_3 and ε_4, substituting from (3.34) into (3.27), Taylor expanding about $h(-u_f^c(0; \varphi_1), z(\varphi_1), u_f^{uu}(0; \varphi_1)) = h(\varphi_1, z(\varphi_1), 0)$, and using (3.32), we have

$$\begin{aligned}h(\varphi_1, z(\varphi_1), 0) + \varepsilon_3 + \mathrm{O}(e^{-\eta L}) = h(\varphi_1, z(\varphi_1), 0) + h_{v^c}(\varphi_1, z(\varphi_1), 0)((\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L})) \\ + h_{v^{ss}}(\varphi_1, z(\varphi_1), 0)(\mathrm{O}(e^{-\eta L})) + \mathrm{O}((\varepsilon_2 - \varepsilon_1 + \mathrm{O}(e^{-\eta L}))^2) + \mathrm{O}(e^{-2\eta L}) \quad (3.39)\end{aligned}$$

so that $\varepsilon_3 = \mathrm{O}(e^{-\eta L})$. By an analogous argument, $\varepsilon_4 = \mathrm{O}(e^{-\eta L})$. Thus $u_1(x)$ with $L, \varphi, w^{ss}, w^{uu}$ as in (3.30)–(3.33) satisfies (3.24), and since $\varepsilon_i = \mathrm{O}(e^{-\eta L})$, $i = 1, 2, 3, 4$, the expressions in (3.14) also hold.

Now (3.14) combined with (3.10)–(3.11) shows $|u_1(-L) - u_f(0; \varphi_1)| = \mathrm{O}(e^{-\eta L})$ and $|u_1(L) - u_b(0; \varphi_2)| = \mathrm{O}(e^{-\eta L})$. Then the expressions (3.15) and (3.18) follow from the stable manifold theorem, while (3.16) and (3.17) follow from the proof of Lemma 3.3 (see Appendix A and [4], Lemma 6.2). The claims for $u_2(x) = \mathcal{R}u_1(-x)$ follow as before. □

These results can be extended to an appropriate infinite dimensional Banach space, as is necessary to address patterns on the plane which are localized in one direction and periodic in the second. Our results immediately carry over to the infinite dimensional case provided the system is such that we can apply the results developed in [26,31] on exponential dichotomies and the existence of stable and unstable manifolds of the periodic orbits $\gamma(x,\mu)$, foliated by smooth strong stable and strong unstable fibers. We refer to the treatment of the Swift–Hohenberg equation on the plane in §6.3 of [4] for further details.

While we have not explicitly addressed the presence of an additional $\mathbb{Z}_2$ symmetry here, we remark the results carry over from [4] in a straightforward way. As shown there, the presence of an additional $\mathbb{Z}_2$ symmetry κ implies that $z(\varphi)$ will be π-periodic (recall that we have taken the minimal period $l(u)$ of the periodic orbits to be 2π.) To see that this implies the existence of four asymmetric solutions with the same L and φ for a given μ, we take without loss of generality $\varphi_1^*, \varphi_2^* \in [0,\pi)$ and note that $\mu = z(\varphi_1^*) = z(\varphi_2^*)$ implies $\mu = z(\varphi_1^* + \pi) = z(\varphi_2^* + \pi)$ and the solutions corresponding to $\varphi_1^*, \varphi_2^*, m = 0, k = k_0$ will have the same L and φ, up to exponentially small corrections, as $\varphi_1^* + \pi, \varphi_2^* + \pi, m = 1, k = k_0 - 1$. That the four solutions actually have the same L and φ follows by applying κ and invoking uniqueness. We note that the solutions can thus be written $u_1(x)$, $\mathcal{R}u_1(-x)$, $\kappa u_1(x)$, and $\kappa\mathcal{R}u_1(-x)$.

4. Numerical illustrations

We now demonstrate that the appearance of asymmetric patterns is as expected based on the above results, using the planar stripe and spot pattern of the planar cubic-quintic Swift–Hohenberg equation:

$$U_t = -(1 + \partial_x^2 + \partial_y^2)^2 U - \mu U + \nu U^3 - U^5, \quad (x,y) \in \mathbb{R} \times S^1 \tag{4.1}$$

where $S^1 = \mathbb{R}/2L_y\mathbb{Z}$ for some $L_y > 0$. We note that the bifurcation diagram for these patterns was first computed in [3]. We recall that steady-state solutions of (4.1) can be considered as solutions to a spatial dynamical system in $u = (u, u_x, u_{xx}, u_{xxx}) \in Y := H^3(S^1) \times H^2(S^1) \times H^1(S^1) \times L^2(S^1)$.

Numerical simulations were completed in MATLAB, using a modified version of EPCONT, a predecessor of COCO [9], as in [3]. The modifications made to EPCONT include employing the Newton trust-region solver FSOLVE, and projecting out the approximate translation directions in each predictor step. We use spectral differentiation matrices in the periodic y-direction, and centered finite differences in the x-direction. In particular, we used $n_y = 8$ Fourier modes and $n_x = 800$ equidistant points on the domain $[-50,50] \times [0,\pi]$ with Neumann boundary conditions. These computations appeared in [21].

Fig. 6 shows the bifurcation diagram for the stripe and spot pattern of (4.1), along with selected solution profiles along an $\mathcal{R}$-symmetric snaking branch and a cross-connecting branch of asymmetric solutions. The term "cross-connecting" refers to the fact that the solution branch originates in a pitchfork bifurcation from the $\mathcal{R}$-symmetric branch, and terminates at a pitchfork bifurcation at the $\kappa\mathcal{R}$-symmetric branch, where $\kappa : u(y) \mapsto -u(L_y - y)$. The solution profiles are shown with the periodic y-axis in the horizontal direction, and the x-axis, along which localization occurs, in the vertical direction. Solutions 1 and 3 are $\mathcal{R}$-symmetric solutions, while solution 2 is an asymmetric solution, all existing for the same value of μ. We observe that solution 2 matches solution 1 for $x \leq 0$ (the top half of the profiles), while it matches solution 3 for $x \geq 0$ (the bottom half of the profiles).

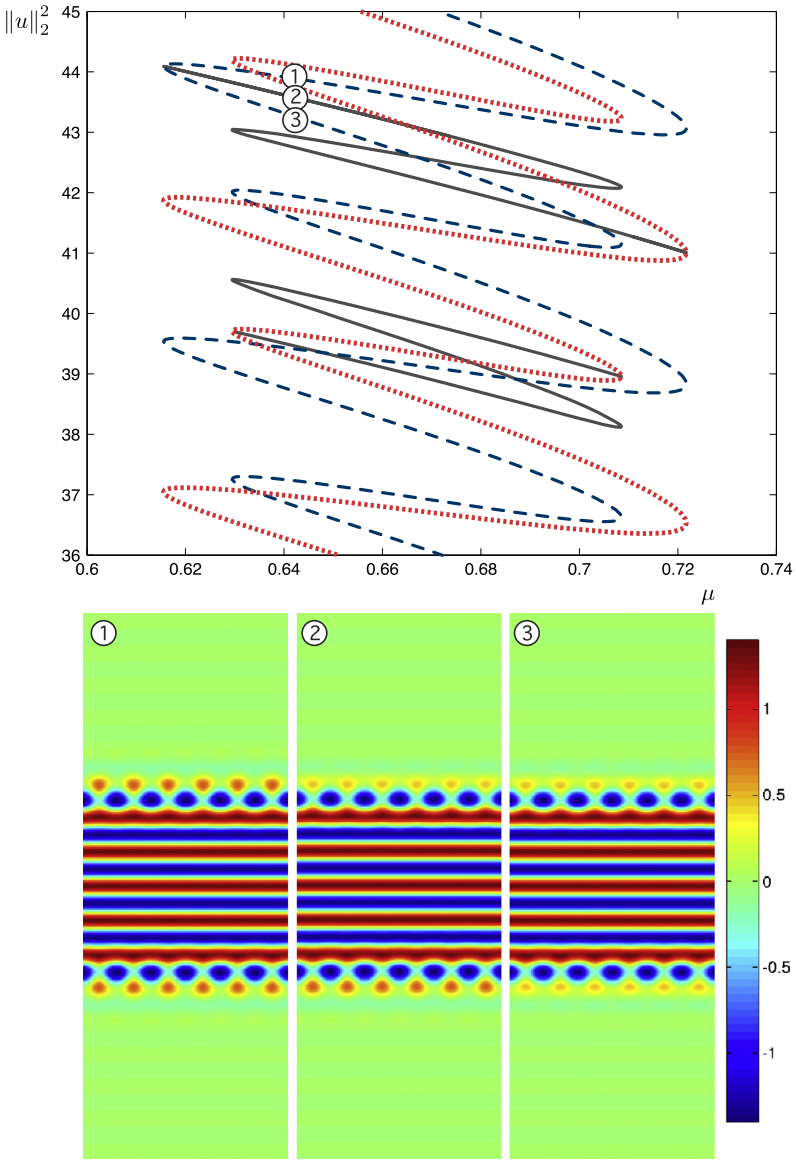


Fig. 6. Bifurcation diagram and particular planar stripes and spots solutions to (4.1), with numbers indicating the correspondence between the bifurcation diagram and solution profiles. In the bifurcation diagram, $\mathcal{R}$-symmetric branches are shown in dashed blue, $\kappa\mathcal{R}$-symmetric branches are in dotted orange, and representative asymmetric branches are in solid gray. Solution profiles are oriented with the x-axis pointing up, and the color indicating the height of the solution, as indicated in the color bar. Recalling that solutions are periodic in the y-direction, we show 6 periods for each solution. Solutions 1 and 3 are $\mathcal{R}$-symmetric solutions, while solution 2 is an asymmetric solution at the same parameter value. We observe that solution 2 matches solution 1 for $x \leq 0$, i.e., the bottom half of the profiles, while solution 2 matches solution 3 for $x \geq 0$, i.e., the top half of the profiles. The bifurcation diagram for these patterns was first computed in [3].

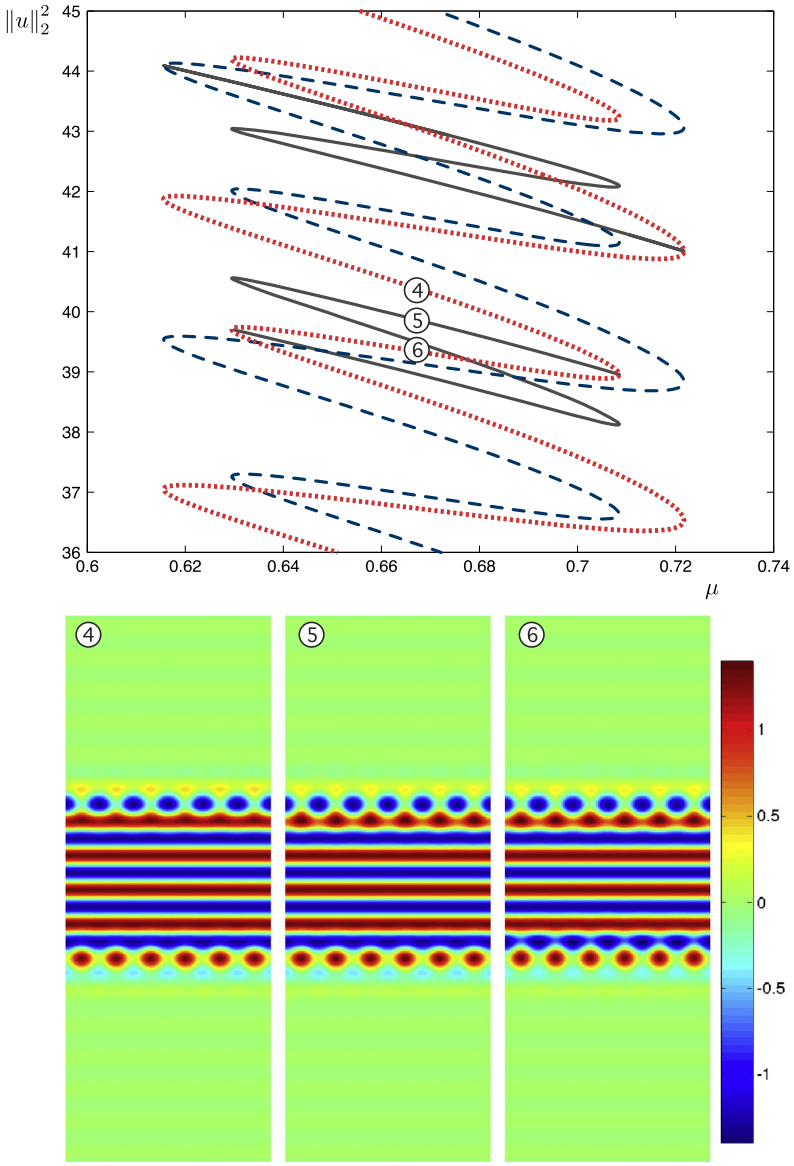


Fig. 7. Bifurcation diagram as in Fig. 6 and additional planar stripes and spots solutions, again with numbers indicating the correspondence between the bifurcation diagram and solution profiles. Here solutions 4 and 6 are $\kappa\mathcal{R}$-symmetric solutions, while solution 5 is an asymmetric solution at the same parameter value. We observe that solution 5 matches solution 4 for $x \leq 0$, i.e., the bottom half of the profiles, while solution 5 matches solution 6 for $x \geq 0$, i.e., the top half of the profiles.

In Fig. 7 we provide an additional illustration of our results. We again show the bifurcation diagram for the stripe and spot pattern of (4.1), along with additional solution profiles, in this case along a $\kappa\mathcal{R}$-symmetric branch and a self-connecting asymmetric branch. A "self-connecting" branch is one that begins and ends at pitchfork bifurcations from the same symmetric branch,

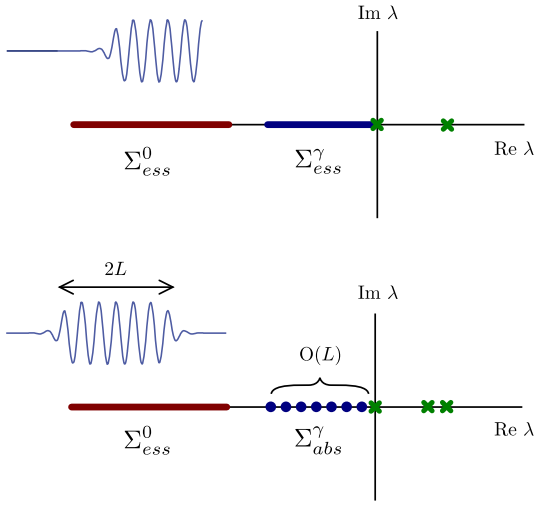


Fig. 8. Schematic diagrams of spectra for the front and pulse solutions. Σ^0_{ess} refers to the essential spectrum of the trivial state, and Σ^γ_{ess} and Σ^γ_{abs} to the essential and absolute spectra of the patterned state. The crosses indicate individual eigenvalues.

in this case a $\kappa\mathcal{R}$-symmetric branch. Here solutions 4 and 6 are $\kappa\mathcal{R}$-symmetric solutions, while solution 5 is an asymmetric solution. Again we observe that solution 5 matches solution 4 for $x \leq 0$ and solution 6 for $x \geq 0$.

5. Overview of stability results

We now turn to the question of stability for localized patterns. Stability is, of course, an important consideration if we hope to be able to say which solutions are experimentally relevant or physically observable in the wide range of natural systems mentioned at the outset. Several questions related to the spectra of localized solutions and fronts are currently outstanding. In Fig. 8, we have plotted the expected spectra of the PDE operators linearized about a front and localized solution, respectively. We emphasize that the bottom diagram depicts our expectation, but remains to be rigorously established under suitable hypotheses in the following.

We first make some remarks on the spectral plots in Fig. 8. The fact that the essential spectrum of a localized solution biasymptotic to the trivial state will correspond to Σ^0_{ess}, the essential spectrum of the trivial state, is well known and we refer to the review article [29]. The appearance of the spectrum for localized solutions in the left half plane, in which the essential spectrum of the periodic state, Σ^γ_{ess}, breaks up into $\mathrm{O}(L)$ eigenvalues has been explained in [30]. As to the eigenvalues with $\mathrm{Re}\,\lambda \geq 0$, a frequently used result establishing the nonlinear stability of traveling wave solutions is that, loosely stated, spectral stability implies nonlinear asymptotic stability provided that the spectrum is bounded away from 0 in the left half plane, with the exception of a simple eigenvalue at $\lambda = 0$ [11]. The application of the result in this case is somewhat delicate due to the accumulation of spectrum up to the imaginary axis as $L \to \infty$, but in any event, the absence of spectrum in the right half plane is a necessary condition for stability.

In the following, we demonstrate exponential closeness of eigenvalues of the localized solution to those of the front and back in the right half plane. Our methods rely on exponential dichotomies, and we must employ different tools once we reach $\lambda = 0$. In a companion paper

[22], we construct an analytic extension of the Evans function of the fronts (and backs) into the left half plane. In this way, we are able to show that provided the front and back have simple eigenvalues at $\lambda = 0$ in an appropriate sense, the eigenvalue of the localized solution at $\lambda = 0$ will also be simple. This is in contrast to the case for localized solutions constructed from gluing together fronts and backs connecting stable rest points (in the PDE sense), where two eigenvalues arise near $\lambda = 0$ [1,25,28].

5.1. General approach

For the following results, we do not require the full set of hypotheses used for the existence results given above. We do, however, wish to use the results on the exponential closeness of the localized solutions $u_1(x)$ to the fronts and backs $u_f(x)$ and $u_b(x)$ and in turn to the periodic solution $\gamma(x)$, as captured in Theorems 1 and 2. In order to clearly identify the hypotheses that are necessary for our results, and to avoid introducing additional assumptions or machinery, we reformulate the results of Theorem 2 and introduce them as Hypotheses 6.1 and 6.2. We explicitly show how these Hypotheses are satisfied in the context of Section 3 in Lemma 6.4.

In the following, we will consider a localized solution along with the front and back to which it is exponentially close, and show that its temporal eigenvalues in the right half plane are exponentially close to those of the front and back, added with multiplicity.

We consider

$$u_x = f(u, \mu) \tag{5.1}$$

with $u \in \mathbb{R}^{2n}$ and $\mu \in \mathbb{R}$.

As described for the particular example of the Swift–Hohenberg equation in Section 1.1, we will be interested in systems of the form

$$\dot{v} = [f_u(u_*(x), \mu) + \lambda B(u_*(x), \mu)]v \tag{5.2}$$

with $v \in \mathbb{C}^{2n}$, $\lambda \in \mathbb{C}$ and $B : \mathbb{R}^{2n} \times \mathbb{R} \to \mathbb{R}^{2n\times 2n}$, and where $u_*(x)$ is a particular solution of (5.1). We assume that f and B are smooth (at least C^2 and C^1, respectively).

We will make use of exponential dichotomies possessed by several systems of the form (5.2), as in Appendix A; we refer to [29] for the statement of the roughness theorem for exponential dichotomies and results on the relationship between the spectrum of a PDE operator, exponential dichotomies of the corresponding ODE, and the Fredholm properties of the family of operators $\mathcal{T}(\lambda)$.

The general idea in the following will be to use information about (5.2) with $u_*(x) = u_f(x), u_b(x)$ or $\gamma(x)$ and the trivial 0 solution at $\lambda = \lambda_*$ to establish the presence or absence of nontrivial, bounded solutions of (5.2) with $u_*(x) = u_\ell(x)$, with $u_\ell(x)$ a localized solution, for λ near λ_*. There are (at least) two possible ways in which to proceed:

(i) Write the solution to the eigenvalue problem (5.2) with $u_*(x) = u_\ell(x)$ for λ near λ_* in terms of the exponential dichotomies for u_f and u_b at $\lambda = \lambda_*$, using the variation of constants formula. As in Remark 6.11, assuming a simple eigenvalue of the front at λ_*, this yields more precise information about the eigenvalue location and eigenfunction form as a perturbation of the eigenfunction of the front; however, at least in the context of determining when a localized solution is stable, this level of detail is not necessary.

(ii) Write the solution to the eigenvalue problem (5.2) with $u_*(x) = u_\ell(x)$ for λ near λ_* directly in terms of exponential dichotomies for this system depending smoothly on λ sufficiently near λ_*, which exist due to the roughness theorem for exponential dichotomies. This is the approach we take in Section 6 to study eigenvalues of arbitrary multiplicity in the right half plane. Since we care primarily about the presence or absence of eigenvalues in the right half plane, rather than precise locations or characterization of the corresponding eigenfunctions, this approach provides shorter and cleaner results.

6. Eigenvalues in the right half plane

We note that in the following we refer to various constants and rates by $C, \eta > 0$ without keeping track of them individually. All such constants may be chosen independently of L for $L > L_*$ sufficiently large and other relevant variables. In particular, we do not distinguish the rates of exponential convergence.

We let $\Omega = \{\lambda \in \mathbb{C} : \mathrm{Re}\lambda > 0\}$. We fix μ so that the following hypotheses are satisfied, and suppress all μ dependence for the remainder of this section.

As described in Lemma 6.4 below, our first hypothesis allows us to introduce a reformulated version of the results given in Theorem 2 without requiring additional hypotheses that are extraneous to our present purposes.

Hypothesis 6.1. There exists a family of solutions $u_{\ell,L}(x)$ to (5.1), parameterized by $L = L(k)$, $k \in \mathbb{N}$, with $k \geq K$ for some $K > 0$, and such that $L(k)$ increases monotonically without bound. These solutions, which we will refer to as localized solutions, satisfy:

$$u_{\ell,L}(x) = \begin{cases} u_f(x+L) + w_f^-(x+L;L), & x \leq -L \\ u_f(x+L) + w_f^+(x+L;L), & x \in [-L, 0] \\ u_b(x-L) + w_b^-(x-L;L), & x \in [0, L] \\ u_b(x-L) + w_b^+(x-L;L), & x \geq L \end{cases} \tag{6.1}$$

where $u_f(x)$ and $u_b(x)$ are also solutions to (5.1) that are independent of L and satisfy $\lim_{x\to-\infty} u_f(x) = 0$ and $\lim_{x\to\infty} u_b(x) = 0$. We also have

$$w_f^-(0;L) = w_f^+(0;L) \tag{6.2}$$

$$u_f(L;L) + w_f^-(L;L) = u_b(-L;L) + w_b^+(-L;L) \tag{6.3}$$

$$w_b^-(0;L) = w_b^+(0;L), \tag{6.4}$$

and there exist positive constants C and η such that

$$\left|w_f^-(x;L)\right| \leq Ce^{-\eta(L-x)}, \quad x \leq 0 \tag{6.5}$$

$$\left|w_f^+(x;L)\right| \leq Ce^{-\eta L}, \quad x \in [0, L] \tag{6.6}$$

$$\left|w_b^-(x;L)\right| \leq Ce^{-\eta L}, \quad x \in [-L, 0] \tag{6.7}$$

$$\left|w_b^+(x;L)\right| \leq Ce^{-\eta(L+x)}, \quad x \geq 0. \tag{6.8}$$

We then require the solutions $u_f(x)$ and $u_b(x)$ to relate to a single solution $\gamma(x)$ as follows:

Hypothesis 6.2. There exists a solution $\gamma(x)$ of (5.1) and positive constants C and η such that the following holds uniformly in $L = L(k)$, $k \geq K$ as in Hypothesis 6.1:

$$|u_f(L) - \gamma(0)| \leq Ce^{-\eta L} \tag{6.9}$$

$$|u_b(-L) - \gamma(0)| \leq Ce^{-\eta L}. \tag{6.10}$$

Moreover, there exists a constant $c \in [0, 2\pi)$, and positive constants C and η, all independent of L, such that

$$|u_f(x) - \gamma(x + c)| \leq Ce^{-\eta x}, \quad x \geq 0 \tag{6.11}$$

$$|u_b(x) - \gamma(x - c)| \leq Ce^{\eta x}, \quad x \leq 0. \tag{6.12}$$

Finally,

$$|u_f(x)| \leq Ce^{\eta x}, \quad x \leq 0 \tag{6.13}$$

$$|u_b(x)| \leq Ce^{-\eta x}, \quad x \geq 0. \tag{6.14}$$

We will use the exponential closeness of $u_f(L)$ and $u_b(-L)$ to $\gamma(0)$ in finding bounded solutions to (5.2) with $u_*(x) = u_{\ell,L}(x)$. We include the exponential approach so that we may take as our final two hypotheses the existence of exponential dichotomies associated to $\gamma(x)$ and the trivial state, from which we deduce the existence of exponential dichotomies of the front and back.

Remark 6.3. Since we no longer have dependence on φ_1 and φ_2, we emphasize that the solutions $u_f(x)$ and $u_b(x)$ are not necessarily related by reversibility, though this possibility is not excluded. Indeed, we do not use or assume reversibility anywhere in this section.

As mentioned already, these hypotheses are in fact satisfied in the set-up used for our existence results:

Lemma 6.4. *Suppose that we are in the setting of Section 3, so that Hypotheses 3.1–3.2 and the analogues of Hypotheses 2.1–2.3 are satisfied, and Theorem 2 holds. Denote by $\hat{\gamma}(x)$, $\hat{u}_f(x)$ and $\hat{u}_b(x)$ the periodic, front, and back solutions introduced in Section 3. Fix $\mu \in J$, $\varphi_1^*, \varphi_2^* \in [0, 2\pi)$, and $m \in \{0, 1\}$, with $\mu = z(\varphi_1^*) = z(\varphi_2^*)$ and $|z'(\varphi_1^*)|, |z'(\varphi_2^*)| \neq 0$. Let φ_1, φ_2 and $L(\varphi_1, \varphi_2, k)$ be as defined in Theorem 2, and let $u_1(x; \varphi_1, \varphi_2, k)$ be the localized solution associated to $\hat{u}_f(x; \varphi_1)$, $\hat{u}_b(x; \varphi_2)$ as given by Theorem 2. Further define $\bar{\varphi} = (\varphi_2 - \varphi_1)/2$ and $c = (\varphi_1 + \varphi_2)/2$ mod 2π. Then there exists $K > 0$ such that Hypotheses 6.1 and 6.2 hold, with $L = L(k)$ as given in Theorem 2, $u_{\ell,L}(x) = u_1(x - \bar{\varphi}; \varphi_1, \varphi_2, k)$, $u_f(x) = \hat{u}_f(x - \bar{\varphi}; \varphi_1)$, $u_b(x) = \hat{u}_b(x - \bar{\varphi}; \varphi_2)$, and $\gamma(x) = \hat{\gamma}(x)$.*

Remark 6.5. As is clear from the following proof, the constant $K > 0$ for which Hypotheses 6.1 and 6.2 hold may need to be larger than that for Theorem 2. The rate $\eta > 0$ appearing in (6.5)–(6.8) is the same as that appearing in Theorem 2, and the constant C can be chosen

independently of μ, φ_1 and φ_2. The $\eta > 0$ appearing in Hypothesis 6.2 is bounded below by the $\eta > 0$ given in Theorem 2.

Proof of Lemma 6.4. Set $L = L(k)$ as in Theorem 2, define $\bar{\varphi} := (\varphi_2 - \varphi_1)/2$, and let $u_{\ell,L}(x) = u_1(x - \bar{\varphi}; \varphi_1, \varphi_2, k)$, $u_f(x) = \hat{u}_f(x - \bar{\varphi}; \varphi_1)$, $u_b(x) = \hat{u}_b(x - \bar{\varphi}; \varphi_2)$. Since φ_1 and φ_2 are fixed, we generally suppress dependence on φ_1 and φ_2 going forward.

In order for (6.1) to hold, we define

$$w_f^-(x; L) := u_{\ell,L}(x - L) - u_f(x) = u_1(x - L - \bar{\varphi}; k) - \hat{u}_f(x - \bar{\varphi}) \tag{6.15}$$

and make analogous definitions for $w_f^+(x; L)$ and $w_b^\pm(x; L)$. Then (6.2)–(6.4) are automatically satisfied due to the continuity of u_1. To avoid cluttered notation in the following, we write $w_{f,b}^\pm(x) = w_{f,b}^\pm(x; L)$ and $u_1(x) = u_1(x; k)$, but dependence on k is, of course, important.

We now consider (6.5)–(6.8). First suppose that $\bar{\varphi} > 0$; we can address the case $\bar{\varphi} < 0$ by reversing the roles of u_f and u_b in the following.

By (3.15) in Theorem 2, we have

$$|u_1(x - L) - \hat{u}_f(x)| \le Ce^{-\eta L}e^{\eta x}, \quad x \le 0 \tag{6.16}$$

so that, since $\bar{\varphi} > 0$,

$$|w_f^-(x)| = |u_1(x - L - \bar{\varphi}) - \hat{u}_f(x - \bar{\varphi})| \le Ce^{-\eta L}e^{\eta x}, \quad x \le 0 \tag{6.17}$$

also holds. Moreover, since $\bar{\varphi} \le 2\pi$, it follows that

$$|w_f^+(x)| = |u_1(x - L - \bar{\varphi}) - \hat{u}_f(x - \bar{\varphi})| \le Ce^{-\eta L}, \quad 0 \le x \le \bar{\varphi}. \tag{6.18}$$

Then from (3.16) in Theorem 2, we have

$$|u_1(x - L) - \hat{u}_f(x)| \le Ce^{-\eta L}, \quad 0 \le x \le L \tag{6.19}$$

so that

$$|w_f^+(x)| = |u_1(x - L - \bar{\varphi}) - \hat{u}_f(x - \bar{\varphi})| \le Ce^{-\eta L}, \quad \bar{\varphi} \le x \le L + \bar{\varphi} \tag{6.20}$$

and, as we will use momentarily,

$$|u_1(x - L - \bar{\varphi}) - \hat{u}_f(x - \bar{\varphi})| \le Ce^{-\eta L}, \quad L \le x \le L + \bar{\varphi}. \tag{6.21}$$

Thus we have shown (6.5) and (6.6) hold. Now from (3.17) in Theorem 2 we have

$$|u_1(x + L) - \hat{u}_b(x)| \le Ce^{-\eta L}, \quad -L \le x \le 0 \tag{6.22}$$

so that

$$|u_1(x + L - \bar{\varphi}) - \hat{u}_b(x - \bar{\varphi})| \le Ce^{-\eta L}, \quad -L + \varphi \le x \le \bar{\varphi} \tag{6.23}$$

To address $|w_b^-(x)|$ on $-L \le x \le -L+\bar{\varphi}$, we note

$$|u_1(x+L-\bar{\varphi}) - \hat{u}_b(x-\bar{\varphi})| \le |u_1(x+L-\bar{\varphi}) - \hat{u}_f(x+2L-\bar{\varphi})| + |\hat{u}_f(x+2L-\bar{\varphi}) - \gamma(x+L)| + |\gamma(x+L) - \hat{u}_b(x-\bar{\varphi})|. \tag{6.24}$$

By (6.21), the first term on the right hand side is bounded by $Ce^{-\eta L}$ for $x \in [-L, -L+\bar{\varphi}]$. We now recall (2.10) and (2.11) from our existence results:

$$|\hat{u}_f(x+\varphi_1) - \gamma(x)| \le Ce^{-\eta(x+\varphi_1)}, \quad x \ge -\varphi_1 \tag{6.25}$$

$$|\hat{u}_b(x-\varphi_2) - \gamma(x)| \le Ce^{\eta(x-\varphi_2)}, \quad x \le \varphi_2 \tag{6.26}$$

where we have reformulated the relations slightly, and stated them for φ_1 and φ_2 for clarity. The second term in (6.24) is equivalent to $|\hat{u}_f(x+L-\bar{\varphi}) - \gamma(x)|$ on $0 \le x \le \bar{\varphi}$, and since $L - \bar{\varphi} = \varphi_1 + 2k\pi + \mathrm{O}(e^{-\eta 2k\pi})$ and $\gamma(x) = \gamma(x+2k\pi)$, we can rewrite this as

$$|\hat{u}_f(x+\varphi_1+2k\pi+\mathrm{O}(e^{-\eta 2k\pi})) - \gamma(x+2k\pi)| \tag{6.27}$$

Then Taylor expanding $\hat{u}_f(x+\varphi_1+2k\pi+\mathrm{O}(e^{-\eta 2k\pi}))$ about $\hat{u}_f(x+\varphi_1+2k\pi)$, we note that $|\hat{u}_f'(x+\varphi_1+2k\pi)|$ is, for any φ_1, uniformly bounded in $k \ge K$, $K > 0$ sufficiently large, by a multiple of $\sup_{x\in[0,2\pi]} |\gamma'(x)|$. So we conclude that

$$|\hat{u}_f(x+L-\bar{\varphi}) - \gamma(x)| \le Ce^{-\eta 2k\pi} \le Ce^{-\eta L} \tag{6.28}$$

uniformly in $L = L(k)$, $k \ge K$ with $K > 0$ sufficiently large. Similarly, the last term in (6.24) is equivalent to $|\gamma(x) - \hat{u}_b(x-L-\bar{\varphi})|$ on $0 \le x \le \bar{\varphi}$ and we note $\bar{\varphi} \le \varphi_2$. We then have $-L-\bar{\varphi} = -\varphi_2 - 2k\pi - \mathrm{O}(e^{-\eta 2k\pi})$ and by an analogous argument as above

$$|\hat{u}_f(x-L-\bar{\varphi}) - \gamma(x)| \le Ce^{-\eta L}. \tag{6.29}$$

This then shows that (6.7) holds. As for (6.8), from (6.23) and $\bar{\varphi} \le 2\pi$ so that we can subsume $e^{\eta 2\pi}$ into the constant, we have

$$|u_1(x+L-\bar{\varphi}) - \hat{u}_b(x-\bar{\varphi})| \le Ce^{-\eta(L+x)}, \quad 0 \le x \le \bar{\varphi} \tag{6.30}$$

Moreover, from (3.18) in Theorem 2 we have

$$|u_1(x+L) - \hat{u}_b(x)| \le Ce^{-\eta(L+x)}, \quad x \ge 0 \tag{6.31}$$

and again since $\bar{\varphi} \le 2\pi$, we have

$$|u_1(x+L-\bar{\varphi}) - \hat{u}_b(x-\bar{\varphi})| \le Ce^{-\eta(L+x)}, \quad x \ge \bar{\varphi}. \tag{6.32}$$

As we remarked at the beginning, the preceding assumed that $\bar{\varphi} > 0$. If $\bar{\varphi} < 0$, we can reverse the arguments, starting with $|w_b^+(x)|$. Thus we conclude that $|w_b^+(x)| \le Ce^{-\eta(L+x)}$ on $x \ge 0$ directly

from (3.18) in Theorem 2, as we did above for $|w_f^-(x)|$ on $x \leq 0$ using (3.15) in Theorem 2. For $|w_b^-(x)|$, we use (3.18) and (3.17) in Theorem 2 in an analogous fashion to the argument above for $|w_f^+(x)|$, and so on. If $\bar{\varphi} = 0$, as is the case for a symmetric localized solution, all the bounds follow immediately from Theorem 2 and the definition (6.15).

Finally, we show that Hypothesis 6.2 holds: we write

$$|u_f(L) - \gamma(0)| \leq |u_f(L) - u_f(2k\pi + \varphi_1 + \bar{\varphi})| + |u_f(2k\pi + \varphi_1 + \bar{\varphi}) - \gamma(0)| \tag{6.33}$$

First considering the first term on the right hand side, by definition $2k\pi + \varphi_1 + \bar{\varphi} = L + \mathrm{O}(e^{-\eta 2k\pi})$, so we can Taylor expand about L and again note that $|u_f'(L)|$ is bounded uniformly in L for any φ_1 by a multiple of $\sup_{x\in[0,2\pi]} |\gamma'(x)|$. Thus the first term is $\mathrm{O}(e^{-\eta 2k\pi}) = \mathrm{O}(e^{-\eta L})$. As for the second term, $u_f(2k\pi + \varphi_1 + \bar{\varphi}) = \hat{u}_f(2k\pi + \varphi_1)$ and $\gamma(0) = \gamma(2k\pi)$, so that by (2.10), the second term is also $\mathrm{O}(e^{-\eta L})$. Thus we have that for $C, \eta > 0$, $|u_f(L) - \gamma(0)| \leq Ce^{-\eta L}$ uniformly in $L = L(k)$, $k \geq K$ with $K > 0$ sufficiently large. To show that (6.10) holds, we write

$$|u_b(-L) - \gamma(0)| \leq |u_b(-L) - u_b(-2k\pi - \varphi_2 + \bar{\varphi})| + |u_b(-2k\pi - \varphi_2 + \bar{\varphi}) - \gamma(0)| \tag{6.34}$$

and argue analogously. Finally, the remaining statements in Hypothesis 6.2 follow from the definition of $u_f(x)$ and $u_b(x)$ upon setting $\tilde{c} = (\varphi_1 + \varphi_2)/2$. As $\tilde{c}$ is bounded and $\gamma(x)$ is 2π periodic, we can then replace $\tilde{c}$ with $c = (\varphi_1 + \varphi_2)/2 \mod 2\pi$. □

We now have two assumptions of the existence of exponential dichotomies for the eigenvalue problems associated to $\gamma(x)$, and the trivial solution at 0. These then imply exponential dichotomies for $u_{f,b}(x)$ on $\mathbb{R}^+$ and $\mathbb{R}^-$, which in turn imply exponential dichotomies for the localized solution on $(-\infty, -L]$, $[-L, 0]$, $[0, L]$ and $[L, \infty)$. As noted below, these are natural assumptions for the PDE formulations we have in mind, but they of course depend on the form of the operator B in (5.2), and do not arise automatically from the set-up in §3. (Although inasmuch as the $\lambda = 0$ case reduces to the problem studied in §3, we note that these hypotheses are compatible with our previous hypotheses on the spatial eigenvalues of the linearizations about the 0 solution and $\gamma(x)$.)

We first assume the existence of an exponential dichotomy of the eigenvalue problem (5.2) with $u_*(x) = \gamma(x)$.

Hypothesis 6.6. The solution $\gamma(x)$ of (5.1) is such that for any $\lambda \in \Omega$,

$$\dot{v} = [f_u(\gamma(x)) + \lambda B(\gamma(x))]v \tag{6.35}$$

has an exponential dichotomy on $\mathbb{R}$. We denote these exponential dichotomies by $\Phi_\gamma^{s,u}(x, y; \lambda)$, and denote the associated projections by $P_\gamma^{s,u}(x; \lambda) = \Phi_\gamma^{s,u}(x, x; \lambda)$. The x-independent dimension $\dim \mathrm{Rg}(P_\gamma^u(x, \lambda))$ is, for all $\lambda \in \Omega$, assumed to be equal to n, so that $\dim \mathrm{Rg}(P_\gamma^s(x, \lambda)) = n$ as well.

Note that in the present context, our only assumptions on $\gamma(x)$ are given in Hypotheses 6.2 and 6.6, and we do not use the particular assumptions on $\gamma(x)$ given in Hypothesis 3.1 above.

Hypothesis 6.6 corresponds to the assumption that the operator corresponding to the linearization about the periodic solution $\gamma(x)$ has no essential spectrum in the open right half plane; as a periodic solution, it has no point spectrum. The condition $\dim \mathrm{Rg}(P^u_\gamma(x,\lambda)) = \dim \mathrm{Rg}(P^s_\gamma(x,\lambda)) = n$ is always satisfied for a reversible system, but see also Remark 6.8.

We finally require that the trivial solution has an exponential dichotomy on $\mathbb{R}$:

Hypothesis 6.7. There exists a constant $d > 0$ such that

$$\dot{v} = [f_u(0) + \lambda B(0)]v \tag{6.36}$$

has an exponential dichotomy on $\mathbb{R}$ for all λ with $\mathrm{Re}\,\lambda > -d$. We denote these dichotomies by $\Phi_0^{s,u}(x,y;\lambda)$, and denote the associated projections by $P_0^{s,u}(x;\lambda)$. The x-independent dimension $\dim \mathrm{Rg}(P_0^u(x,\lambda))$ is, for all λ with $\mathrm{Re}\,\lambda > -d$, assumed to be equal to n, so that $\dim \mathrm{Rg}(P_0^s(x,\lambda)) = n$ as well.

Hypothesis 6.7 corresponds to the assumption that the essential spectrum of the operator corresponding to the linearization about the trivial solution is bounded away from 0 in the left half plane. For our immediate purposes we only need the existence of exponential dichotomies for $\lambda \in \Omega$, the open right half plane, but we would ultimately need this stronger condition to show the eigenvalue at $\lambda = 0$ remains simple, as we do in [22].

Taken together, Hypotheses 6.6 and 6.7, along with Hypothesis 6.2 and the roughness theorem for exponential dichotomies, then imply that the front and back solutions have exponential dichotomies on $\mathbb{R}^\pm$. More precisely, for any $\lambda \in \Omega$,

$$\dot{v} = [f_u(u_j(x)) + \lambda B(u_j(x))]v \tag{6.37}$$

with $j = f, b$ has exponential dichotomies on $\mathbb{R}^+$ and $\mathbb{R}^-$, which we can always choose to be locally analytic. We denote these dichotomies by $\Phi^{s,u}_{j,+}(x,y;\lambda)$ and $\Phi^{s,u}_{j,-}(x,y;\lambda)$ with $j = f, b$, where the subscripts $+$ and $-$ correspond to the dichotomies on $\mathbb{R}^+$ and $\mathbb{R}^-$, respectively. We again denote the associated projections by $P^{s,u}_{j,\pm}(x;\lambda) := \Phi^{s,u}_{j,\pm}(x,x;\lambda)$. The dimension of the range of each of these projections is equal to n. This indicates that the operators corresponding to the linearizations about the front and back solutions have no essential spectrum in the open right half plane, but only isolated eigenvalues.

Remark 6.8. In a general system it may happen that the range of the projection operator associated to each exponential dichotomy has dimension other than n, in contrast to the situation for a reversible system where this dimension must equal n. To account for this possibility, in the above hypotheses we could instead allow an arbitrary dimension, $\dim \mathrm{Rg}(P^u_\gamma(x,\lambda)) = i_\infty$, so that $\dim \mathrm{Rg}(P^s_\gamma(x,\lambda)) = 2n - i_\infty$. The dimension i_∞ is commonly referred to as the Morse index. If it happened that $i_\infty \neq n$, we would then have the additional requirement that the dimensions of the exponential dichotomy associated to the trivial solution is compatible with that of γ:

$$\dim \mathrm{Rg}(P^u_0(x,\lambda)) = \dim \mathrm{Rg}(P^u_\gamma(x,\lambda)) = i_\infty$$

which would of course imply

$$\dim \mathrm{Rg}(P^s_0(x,\lambda)) = \dim \mathrm{Rg}(P^s_\gamma(x,\lambda)) = 2n - i_\infty.$$

This would then imply that the dimensions of the dichotomies associated with u_f and u_b on $\mathbb{R}^{\pm}$ are compatible with each other and with the dichotomies associated to γ and the trivial solution. Our results are the same whether we allow an arbitrary dimension i_∞ and enforce the above compatibility condition, or take the dimension of the range of all projection operators to be n; for simplicity, we make the latter choice.

We are now in a position to state the main result of this section:

Theorem 3. *Suppose Hypotheses 6.1, 6.2, 6.6 and 6.7 are satisfied, and define the functions*

$$D_f(\lambda) := \det\left(\operatorname{Rg} P^u_{f,-}(0;\lambda), \operatorname{Rg} P^s_{f,+}(0;\lambda)\right) \tag{6.38}$$

$$D_b(\lambda) := \det\left(\operatorname{Rg} P^u_{b,-}(0;\lambda), \operatorname{Rg} P^s_{b,+}(0;\lambda)\right). \tag{6.39}$$

Fix $\lambda_* \in \Omega$ and suppose that for $m_f, m_b \geq 0$ and for some $\delta > 0$,

$$D_f(\lambda) = (\lambda - \lambda_*)^{m_f} + \mathrm{O}(|\lambda - \lambda_*|^{m_f+1}) \tag{6.40}$$

and

$$D_b(\lambda) = (\lambda - \lambda_*)^{m_b} + \mathrm{O}(|\lambda - \lambda_*|^{m_b+1}) \tag{6.41}$$

for $\lambda \in U_\delta(\lambda_*)$, the ball of radius δ centered at λ_*. Then we may define an analytic function $D_{\ell,L}(\lambda)$ such that there exists a $\hat{\delta}$ sufficiently small, with $\delta > \hat{\delta} > 0$, and an $L_* > 0$ sufficiently large, such that the following hold uniformly in $L \geq L_*$:

(i) *$D_{\ell,L}(\lambda)$ has precisely $m_f + m_b$ roots, counted with multiplicity, in $U_{\hat{\delta}}(\lambda_*)$. These values of λ are $\mathrm{O}(e^{-\eta L})$ close to λ_*, with $\eta > 0$.*

(ii) *The system*

$$\dot{v} = \left[f_u(u_{\ell,L}(x)) + \lambda B(u_{\ell,L}(x))\right] v, \tag{6.42}$$

has a bounded, nontrivial solution at $\lambda \in U_{\hat{\delta}}(\lambda_*)$ if and only if $D_{\ell,L}(\lambda) = 0$.

Proof. We will show that we have a bounded, nontrivial solution to (6.42) if and only if $D_{\ell,L}(\lambda) = 0$, where $D_{\ell,L}(\lambda)$ is an analytic function of the form

$$D_{\ell,L}(\lambda) = \left(D_f(\lambda) + \mathrm{O}(e^{-\eta L})\right)\left(D_b(\lambda) + \mathrm{O}(e^{-\eta L})\right) + \mathrm{O}(e^{-2\eta L}). \tag{6.43}$$

We begin by noting that using (6.1), we can solve the eigenvalue problem (6.42) in pieces by solving the equations

$$\dot{v}^-_f = \left[f_u(u_f(x) + w^-_f(x)) + \lambda B(u_f(x) + w^-_f(x))\right] v^-_f, \quad x \leq 0 \tag{6.44}$$

$$\dot{v}^+_f = \left[f_u(u_f(x) + w^+_f(x)) + \lambda B(u_f(x) + w^+_f(x))\right] v^+_f, \quad x \in [0, L] \tag{6.45}$$

$$\dot{v}^-_b = \left[f_u(u_b(x) + w^-_b(x)) + \lambda B(u_b(x) + w^-_b(x))\right] v^-_b, \quad x \in [-L, 0] \tag{6.46}$$

$$\dot{v}_b^+ = \Big[f_u(u_b(x) + w_b^+(x)) + \lambda B(u_b(x) + w_b^+(x)) \Big] v_b^+, \quad x \geq 0 \tag{6.47}$$

and then constructing our solution as

$$v(x) = \begin{cases} v_f^-(x+L), & x \leq -L \\ v_f^+(x+L), & x \in [-L, 0] \\ v_b^-(x-L), & x \in [0, L] \\ v_b^+(x-L), & x \geq L. \end{cases} \tag{6.48}$$

Thus we have a bounded solution to (6.42) if and only if we have solutions to (6.44)–(6.47) along with the matching conditions

$$v_f^+(L) - v_b^-(-L) = 0 \tag{6.49}$$

$$v_f^+(0) - v_f^-(0) = 0 \tag{6.50}$$

$$v_b^+(0) - v_b^-(0) = 0. \tag{6.51}$$

We fix $\lambda_* \in \Omega$ as in the Theorem statement. By the existence of exponential dichotomies for (6.37) and the roughness theorem for exponential dichotomies, we have exponential dichotomies for each equation (6.44)–(6.47), which can be chosen so that they depend analytically on $\lambda \in U_\delta(\lambda_*)$ for $\delta > 0$ small. That is, we have

$$\begin{aligned}
&|\Phi^s_{\ell(f,L,-)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \geq y, \quad |\Phi^u_{\ell(f,L,-)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \leq y, \\
&\quad x, y \leq 0 \\
&|\Phi^s_{\ell(f,L,+)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \geq y, \quad |\Phi^u_{\ell(f,L,+)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \leq y, \\
&\quad x, y \in [0, L] \\
&|\Phi^s_{\ell(b,L,-)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \geq y, \quad |\Phi^u_{\ell(b,L,-)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \leq y, \\
&\quad x, y \in [-L, 0] \\
&|\Phi^s_{\ell(b,L,+)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \geq y, \quad |\Phi^u_{\ell(b,L,+)}(x,y;\lambda)| \leq Ce^{-\eta|x-y|},\ x \leq y, \\
&\quad x, y \geq 0
\end{aligned} \tag{6.52}$$

We denote the associated projections $P^{s,u}_{\ell(j,L,\pm)}(x;\lambda) := \Phi^{s,u}_{\ell(j,L,\pm)}(x,x;\lambda)$, $j = f, b$. Moreover, as a consequence of Hypothesis 6.6, we have that

$$|P^u_{\ell(f,L+)}(L;\lambda) - P^u_\gamma(0;\lambda)| \leq Ce^{-\eta L} \tag{6.53}$$

$$|P^s_{\ell(b,L,-)}(-L;\lambda) - P^s_\gamma(0;\lambda)| \leq Ce^{-\eta L}. \tag{6.54}$$

We now let $a := (a_f^+, a_b^-) \in V_a$, $b := (b_f^-, b_f^+, b_b^-, b_b^+) \in V_b$ and $\lambda \in V_\lambda$, where the spaces V_a, V_b, V_λ are defined as follows:

$$
\begin{aligned}
V_a &:= \operatorname{Rg} P^u_\gamma(0;\lambda_*) \oplus \operatorname{Rg} P^s_\gamma(0;\lambda_*)\\
V_b &:= \left(\operatorname{Rg} P^u_{\ell(f,L,-)}(0;\lambda_*) \times \operatorname{Rg} P^s_{\ell(f,L,+)}(0;\lambda_*)\right)\\
&\quad \times \left(\operatorname{Rg} P^u_{\ell(b,L,-)}(0;\lambda_*) \times \operatorname{Rg} P^s_{\ell(b,L,+)}(0;\lambda_*)\right)\\
V_\lambda &:= U_\delta(\lambda_*) \subset \mathbb{C},
\end{aligned}
$$

and where V_a and V_b are endowed with the maximum norm over components.

For δ sufficiently small and $L > L_*$ sufficiently large, we can then write solutions to the eigenvalue problem (6.42) for the localized structure $u_{\ell,L}$ as

$$v^-_f(x) = \Phi^u_{\ell(f,L,-)}(x,0;\lambda)b^-_f \tag{6.55}$$

$$v^+_f(x) = \Phi^s_{\ell(f,L,+)}(x,0;\lambda)b^+_f + \Phi^u_{\ell(f,L,+)}(x,L;\lambda)a^+_f \tag{6.56}$$

$$v^-_b(x) = \Phi^s_{\ell(b,L,-)}(x,-L;\lambda)a^-_b + \Phi^u_{\ell(b,L,-)}(x,0;\lambda)b^-_b \tag{6.57}$$

$$v^+_b(x) = \Phi^s_{\ell(b,L,+)}(x,0;\lambda)b^+_b, \tag{6.58}$$

where these are defined on $(-\infty,0]$, $[0,L]$, $[-L,0]$ and $[0,\infty)$, respectively. Clearly any solution of (6.55)–(6.58) and the matching conditions (6.49)–(6.51) with a and b not both 0 will be a nontrivial, bounded solution to (6.42). On the other hand, all bounded solutions to (6.42) can be found as solutions to (6.55)–(6.58) and (6.49)–(6.51) provided (i) δ is sufficiently small such that $\operatorname{Rg} P^s_{\ell(f,L,-)}(0;\lambda) \cap \operatorname{N} P^s_{\ell(f,L,-)}(0;\lambda_*) = 0$ and similarly for the other projections operating on V_b, which is guaranteed to hold for some $\delta > 0$ by the analytic dependence of the exponential dichotomies on λ near λ_*, and (ii) δ is sufficiently small and L_* is sufficiently large such that $\operatorname{Rg} P^u_{\ell(f,L,+)}(L;\lambda) \cap \operatorname{N} P^u_\gamma(0;\lambda_*) = 0$, and similarly $\operatorname{Rg} P^s_{\ell(b,L,-)}(-L;\lambda) \cap \operatorname{N} P^s_\gamma(0;\lambda_*) = 0$, which holds again by analytic dependence of the exponential dichotomies, along with the relations (6.53) and (6.54).

We now use the matching conditions (6.49)–(6.51) to solve for a and b. We begin by using the first matching condition (6.49) to solve for a in terms of b.

Lemma 6.9. *There exists an L_* such that for all $L > L_*$, the following holds uniformly in L. There exists an operator $A_1 : V_\lambda \times V_b \to V_a$ such that $v(x)$ as given by (6.55)–(6.58) with $a = A_1(\lambda)b$ solves (6.49) for any b and λ. A_1 is analytic in λ and linear in b, and satisfies*

$$|A_1(\lambda)b| \le Ce^{-\eta L}|b|. \tag{6.59}$$

Proof. Substituting from (6.55)–(6.58) into (6.49) gives us:

$$
\begin{aligned}
0 &= \Phi^s_{\ell(f,L,+)}(L,0;\lambda)b^+_f + \Phi^u_{\ell(f,L,+)}(L,L;\lambda)a^+_f\\
&\quad - \Phi^s_{\ell(b,L,-)}(-L,-L;\lambda)a^-_b - \Phi^u_{\ell(b,L,-)}(-L,0;\lambda)b^-_b\\
&= a^+_f - a^-_f + (P^u_{\ell(f,L,+)}(L;\lambda) - P^u_\gamma(0;\lambda))a^+_f + (P^s_\gamma(0;\lambda) - P^s_{\ell(f,L,+)}(L;\lambda))a^-_b\\
&\quad + \Phi^s_{\ell(f,L,+)}(L,0;\lambda)b^+_f - \Phi^u_{\ell(b,L,-)}(-L,0;\lambda)b^-_b.
\end{aligned}
\tag{6.60}
$$

We then define

$$F_1(\lambda)(a,b) := (P^u_{\ell(f,L,+)}(L;\lambda) - P^u_\gamma(0;\lambda))a^+_f + (P^s_\gamma(0;\lambda) - P^s_{\ell(f,L,+)}(L;\lambda))a^-_b + \Phi^s_{\ell(f,L,+)}(L,0;\lambda)b^+_f - \Phi^u_{\ell(b,L,-)}(-L,0;\lambda)b^-_b \tag{6.61}$$

so that clearly $F_1(\lambda)(a,b)$ is analytic in λ and linear in a and b. Moreover, from (6.53) and (6.54) and definition of the dichotomies, i.e., the decay of $\Phi^s_{\ell(f,L,+)}(x,y;\lambda)$ on $x \geq y$ and $\Phi^u_{\ell(b,L,-)}(x,y;\lambda)$ on $x \leq y$, we have

$$|F_1(\lambda)(a,b)| \leq Ce^{-\eta L}(|a| + |b|). \tag{6.62}$$

We further define the map J as

$$\begin{aligned} J : V_a &\to \mathbb{C}^{2n} \\ (a^+_f, a^-_b) &\mapsto a^+_f - a^-_b, \end{aligned} \tag{6.63}$$

and since $V_a = \operatorname{Rg} P^u_\gamma(0;\lambda_*) \oplus \operatorname{Rg} P^s_\gamma(0;\lambda_*) = \mathbb{C}^{2n}$, we have that J is a linear isomorphism and $\|J\| \leq C$. Then we can rewrite (6.60) as

$$(J + F_1(\lambda)I_1)a = F_1(\lambda)(0,b), \tag{6.64}$$

where $I_1 a := (a,0)$. By (6.62), for $L \geq L_*$ with L_* sufficiently large, $(J + F_1(\lambda)I_1)$ is invertible so that we have the solution operator

$$a = -(J + F_1(\lambda)I_1)^{-1}F_1(\lambda)(0,b) =: A_1(\lambda)b. \tag{6.65}$$

Finally, again from (6.62) we have

$$|A_1(\lambda)b| \leq Ce^{-\eta L}|b|, \tag{6.66}$$

which completes the proof. □

Turning now to the second matching condition $v^+_f(0) - v^-_f(0) = 0$, we have

$$\begin{aligned} 0 &= \Phi^s_{\ell(f,L,+)}(0,0;\lambda)b^+_f + \Phi^u_{\ell(f,L,+)}(0,L;\lambda)a^+_f - \Phi^u_{\ell(f,L,-)}(0,0;\lambda)b^-_f \\ &= P^s_{\ell(f,L,+)}(0;\lambda)b^+_f - P^u_{\ell(f,L,-)}(0;\lambda)b^-_f + \Phi^u_{\ell(f,L,+)}(0,L;\lambda)(A_1(\lambda)b)^+_f, \end{aligned} \tag{6.67}$$

where we have defined $(A_1(\lambda)b)^+_f$ to be the component of $A_1(\lambda)b$ in $\operatorname{Rg} P^u_\gamma(0;\lambda_*)$. We observe that by (6.52) and (6.59),

$$\Phi^u_{\ell(f,L,+)}(0,L;\lambda)(B(\lambda)b)^+_f = \mathrm{O}(e^{-2\eta L}|b|) \tag{6.68}$$

uniformly in λ near λ_*.

Similarly, from the third matching condition $v_b^+(0) - v_b^-(0) = 0$, we have

$$\begin{aligned} 0 &= \Phi^s_{\ell(b,L,-)}(0,0;\lambda)b_b^+ - \Phi^u_{\ell(b,L,-)}(0,0;\lambda)b_b^- - \Phi^s_{\ell(b,L,-)}(0,-L;\lambda)(A_1(\lambda)b)_b^- \\ &= P^s_{\ell(b,L,-)}(0;\lambda)b_b^+ - P^u_{(\ell(b,L,-)}(0;\lambda)b_b^- - \Phi^s_{\ell(b,L,-)}(0,-L;\lambda)(A_1(\lambda)b)_b^-, \end{aligned} \tag{6.69}$$

where $(A_1(\lambda)b)_b^-$ is the component of $A_1(\lambda)b$ in $\operatorname{Rg} P^s_\gamma(0;\lambda_*)$. From (6.52) and (6.59), we again have

$$\Phi^s_{\ell(b,L,-)}(0,-L;\lambda)(A_1(\lambda)(b))_b^- = \mathrm{O}(e^{-2\eta L}|b|) \tag{6.70}$$

uniformly in λ near λ_*.

We have a nontrivial solution at λ near λ_* if the conditions (6.67) and (6.69) hold for $b \neq 0$. We can rewrite the two conditions as:

$$\begin{aligned} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \Bigg[&\begin{pmatrix} -P^u_{\ell(f,L,-)}(0;\lambda) & P^s_{\ell(f,L,+)}(0;\lambda) & 0 & 0 \\ 0 & 0 & -P^u_{\ell(b,L,-)}(0;\lambda) & P^s_{\ell(b,L,+)}(0;\lambda) \end{pmatrix} \\ &+ \begin{pmatrix} \Phi^u_{\ell(f,L,+)}(0,L;\lambda) & -\Phi^s_{\ell(b,L,-)}(0,-L;\lambda) \end{pmatrix} A_1(\lambda) \Bigg] b \\ =: &\,[P_L(\lambda) + R_L(\lambda)]\, b. \end{aligned} \tag{6.71}$$

We note that these matrices are square, as we are mapping from V_b with $4n$ complex components (for general Morse indices, $i_\infty + (2n - i_\infty) + i_\infty + (2n - i_\infty) = 4n$) to $\mathbb{C}^{2n} \times \mathbb{C}^{2n}$. From (6.68) and (6.70),

$$R_L(\lambda) = \mathrm{O}(e^{-2\eta L}), \tag{6.72}$$

uniformly in λ near λ_*.

We will then have a nontrivial solution to (6.44)–(6.47) along with the matching conditions (6.49)–(6.51) at λ near λ_* provided that we can find $b \in V_b$, $b \neq 0$ such that $(P_L(\lambda) + R_L(\lambda))b = 0$. Thus we define $\hat{P}^u_{\ell(f,L,-)}(0;\lambda)$ as $P^u_{\ell(f,L,-)}(0;\lambda)$ restricted to $\operatorname{Rg} P^u_{\ell(f,L,-)}(0;\lambda_*)$:

$$\hat{P}^u_{\ell(f,L,-)}(0;\lambda) := P^u_{\ell(f,L,-)}(0;\lambda)\Big|_{\operatorname{Rg} P^u_{\ell(f,L,-)}(0;\lambda_*)} \tag{6.73}$$

and similarly for the other projection operators. We further define

$$\hat{P}_L(\lambda) = \begin{pmatrix} -\hat{P}^u_{\ell(f,L,-)}(0;\lambda) & \hat{P}^s_{\ell(f,L,+)}(0;\lambda) & 0 & 0 \\ 0 & 0 & -\hat{P}^u_{\ell(b,L,-)}(0;\lambda) & \hat{P}^s_{\ell(b,L,+)}(0;\lambda) \end{pmatrix} \tag{6.74}$$

and

$$\hat{R}_L(\lambda) = R_L(\lambda)\Big|_{V_b}. \tag{6.75}$$

We thus define

$$D_{\ell,L}(\lambda) := \det(\hat{P}_L(\lambda) + \hat{R}_L(\lambda)) \tag{6.76}$$

so that we have a nontrivial solution of (6.42) at λ near λ_* if and only if

$$D_{\ell,L}(\lambda) = 0. \tag{6.77}$$

Now since $\hat{P}_L(\lambda)$ and $\hat{R}_L(\lambda)$ are analytic, as is the determinant, we can write this as

$$\det\left(\hat{P}_L(\lambda)\right) + \tilde{R}_L(\lambda) = 0, \tag{6.78}$$

where

$$\tilde{R}_L(\lambda) = \mathrm{O}\left(|\hat{R}_L(\lambda)|\right) = \mathrm{O}(e^{-2\eta L}), \tag{6.79}$$

and we clarify that the first equality is meant asymptotically as $\lambda \to \lambda_*$ and the second as $L \to \infty$, so that taken together there exists a C which does not depend on λ or L such that for $\lambda \in U_\delta$ with δ sufficiently small and $L > L_*$, with L_* sufficiently large,

$$\tilde{R}_L(\lambda) \leq Ce^{-2\eta L}. \tag{6.80}$$

Focusing then on $\det(\hat{P}_L(\lambda))$, we have

$$\begin{aligned}\det(\hat{P}_L(\lambda)) = \det\Big(-\hat{P}^u_{\ell(f,L,-)}(0;\lambda) \quad \hat{P}^s_{\ell(f,L,+)}(0;\lambda) \Big) \\ \times \det\Big(-\hat{P}^u_{(\ell(b,L,-)}(0;\lambda) \quad \hat{P}^s_{\ell(b,L,-)}(0;\lambda) \Big)\end{aligned} \tag{6.81}$$

where again the hats indicate that the projections are restricted to the range of the corresponding projection with $\lambda = \lambda_*$. But we note that

$$\mathrm{Rg}\, P^u_{\ell(f,L,-)}(0;\lambda)\Big|_{\mathrm{Rg}\, P^u_{\ell(f,L,-)}(0;\lambda_*)} \to \mathrm{Rg}\, P^u_{\ell(f,L,-)}(0;\lambda) \tag{6.82}$$

is an isomorphism with uniform bound in λ near λ_*, and similarly for the other projection operators. So then we have a nontrivial solution if and only if

$$\begin{aligned}\det\Big(\mathrm{Rg}\, P^u_{\ell(f,L,-)}(0;\lambda), \mathrm{Rg}\, P^s_{\ell(f,L,+)}(0;\lambda)\Big) \det\Big(\mathrm{Rg}\, P^u_{\ell(b,L,-)}(0;\lambda), \mathrm{Rg}\, P^s_{\ell(b,L,+)}(0;\lambda)\Big) \\ + \tilde{R}_L(\lambda) = 0.\end{aligned} \tag{6.83}$$

Now since we also have that $|P^s_{\ell(f,L,+)}(0;\lambda) - P^s_{f,+}(0;\lambda)| \leq Ce^{-\eta L}$, and analogously for projections associated with the other exponential dichotomies, we have

$$\begin{aligned} 0 = &\left(\det\left(\operatorname{Rg} P^s_{f,+}(0;\lambda), \operatorname{Rg} P^u_{f,-}(0;\lambda)\right) + \mathrm{O}(e^{-\eta L})\right) \\ &\times \left(\det\left(\operatorname{Rg} P^s_{b,-}(0;\lambda), \operatorname{Rg} P^u_{b,+}(0;\lambda)\right) + \mathrm{O}(e^{-\eta L})\right) \\ &+ \mathrm{O}(e^{-2\eta L}). \end{aligned} \tag{6.84}$$

We note that by the definitions (6.40) and (6.41), our requirement can then be written

$$D_\ell(\lambda) = \big(D_f(\lambda) + \mathrm{O}(e^{-\eta L})\big)\big(D_b(\lambda) + \mathrm{O}(e^{-\eta L})\big) + \mathrm{O}(e^{-2\eta L}) = 0. \tag{6.85}$$

Since, by assumption,

$$D_f(\lambda) = (\lambda - \lambda_*)^{m_f} + \mathrm{O}(|\lambda - \lambda_*|^{m_f+1}) \tag{6.86}$$

and

$$D_b(\lambda) = (\lambda - \lambda_*)^{m_b} + \mathrm{O}(|\lambda - \lambda_*|^{m_b+1}) \tag{6.87}$$

hold for $\lambda \in U_\delta(\lambda_*)$, we can use Rouché's Theorem to complete our proof:

Lemma 6.10. *Suppose that* (6.86) *and* (6.87) *hold for* $\lambda \in U_{\delta_0}(\lambda_*)$ *and* $D_{\ell,L}(\lambda)$ *is as defined in* (6.76) *for* $\lambda \in U_{\delta_1}(\lambda_*)$. *Then there exists a* $\hat\delta$ *satisfying* $\min\{\delta_0, \delta_1\} > \hat\delta > 0$ *such that for* $L_* > 0$ *sufficiently large,* $D_{\ell,L}(\lambda)$ *has exactly* $m_f + m_b$ *zeroes in* $U_{\hat\delta}(\lambda_*)$, *uniformly in* $L \geq L_*$, *which are* $\mathrm{O}(e^{-\eta L/f(m_f,m_b)})$ *close to* λ_*, *with* $f(m_f, m_b) = \max\{m_f, m_b\} + \epsilon$ *and* $\epsilon > 0$ *arbitrarily small.*

Proof. Rouché's theorem can be stated as follows: Let $K \subset \mathbb{C}$ a closed, bounded region with ∂K a simple closed contour, and suppose $f, g : K \to \mathbb{C}$ are analytic with

$$|f(z) - g(z)| < |f(z)| + |g(z)| \tag{6.88}$$

on ∂K. Then f and g have the same number of zeros in K, counted with multiplicity.

In the present case, working from the expression for $D_{\ell,L}(\lambda)$ given in (6.85), we let $z = \lambda - \lambda_*$, and define

$$\begin{aligned} f(z) &= z^{m_f+m_b} \\ g(z) &= z^{m_f+m_b} + \mathrm{O}(z^{m_f+m_b+1}) + \mathrm{O}(e^{-\eta L}(z^{m_f} + z^{m_b})) + \mathrm{O}(e^{-2\eta L}) \end{aligned} \tag{6.89}$$

with $K = B_r(0)$ for some $r > 0$. We wish to find the smallest r such that f and g have the same number of zeros in K. We require

$$\begin{aligned} \mathrm{O}(r^{m_f+m_b+1}) + \mathrm{O}(e^{-\eta L}(r^{m_f} + r^{m_b})) + \mathrm{O}(e^{-2\eta L}) < 2r^{m_f+m_b} - \mathrm{O}(r^{m_f+m_b+1}) \\ - \mathrm{O}(e^{-\eta L}(r^{m_f} + r^{m_b})) - \mathrm{O}(e^{-2\eta L}) \end{aligned} \tag{6.90}$$

or

$$\mathrm{O}(e^{-2\eta L}) < 2r^{m_f+m_b}\left(1-\mathrm{O}(r)-\mathrm{O}\left(e^{-\eta L}(r^{-m_b}+r^{-m_f})\right)\right) \tag{6.91}$$

In other words, for L_* sufficiently large, there exists a C such that we can write the requirement as

$$Ce^{-2\eta L} < 2r^{m_f+m_b}\left(1-Cr-Ce^{-\eta L}(r^{-m_b}+r^{-m_f})\right). \tag{6.92}$$

Then setting $r = C^{-1}e^{-\eta L/f(m_f,m_b)}$, where $f(m_f,m_b)$ is a function to be determined, we require $\frac{m_f+m_b}{f(m_f,m_b)} \leq 1$ as well as $\frac{m_f}{f(m_f,m_b)} < \frac{1}{2}$ and $\frac{m_b}{f(m_f,m_b)} < \frac{1}{2}$, where we require strict inequalities to ensure the second factor in (6.92) can be made arbitrarily close to 1 without requiring dependence of the constant on m_f or m_b. Thus we need $f(m_f,m_b) > 2\max\{m_f,m_b\}$, and so we conclude that $r = \mathrm{O}(e^{-\eta L/f(m_f,m_b)})$, where $f(m_f,m_b) = 2\max\{m_f,m_b\}+\epsilon$, where $\epsilon > 0$. □

This completes the proof of Theorem 3. □

Remark 6.11. Additional information about the eigenvalues and eigenfunctions associated to localized solutions may be obtained by following a slightly different but closely related strategy to that employed here. In particular, in the case of a simple eigenvalue of the front we have the following result:

Theorem 4. *There exists an $L_* > 0$ such that the following holds uniformly in $L > L_*$. There exists a small $\delta > 0$ such that (6.42) has a bounded, nonzero solution for $\lambda \in U_\delta(\lambda_*)$ if and only if*

$$-\left\langle \psi_*(L), P^u_\gamma(0;\lambda_*)v_{f*}(L)\right\rangle - (\lambda-\lambda_*)M + R(\lambda-\lambda_*) = 0 \tag{6.93}$$

where

$$M = \int_{-\infty}^{\infty} \left\langle \psi_*(s), B(u_f(s))v_{f*}(s)\right\rangle ds \tag{6.94}$$

and R is analytic in $\lambda-\lambda_*$ and satisfies

$$|R(\lambda-\lambda_*)| \leq C\left(e^{-\alpha L_*}\left(\sup_{s\geq L_*}|B(u_f)(s)v_{f*}(s)| + |v_{f*}(L_*)| + e^{-\eta L_*} + |\lambda-\lambda_*|\right) + e^{-\eta L_*}\right). \tag{6.95}$$

Here $v_{f*}(x)$ is the eigenfunction associated with the eigenvalue λ_* of the front $u_f(x)$ at λ_*, i.e., the unique bounded solution to

$$\dot v = \left[f_u(u_f(x)) + \lambda B(u_f(x))\right]v \tag{6.96}$$

at λ_*, while ψ_* is the unique bounded solution of the adjoint equation

$$\dot w = -\left[f_u(u_f(x)) + \lambda B(u_f(x))\right]^* w \tag{6.97}$$

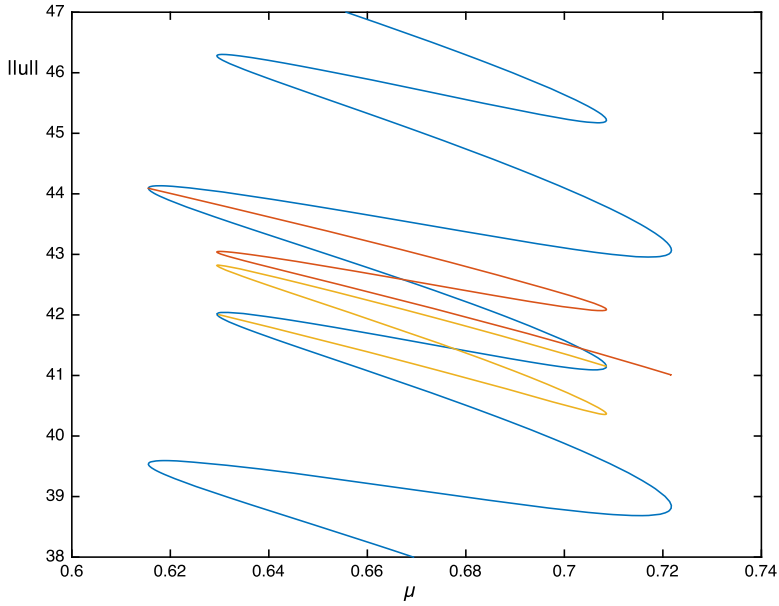


Fig. 9. Selected branches in the bifurcation diagram for stripe and spot patterns, showing the three branches of solutions whose spectra will be analyzed in the following: symmetric (blue), asymmetric cross-connecting (orange) and asymmetric self-connecting (yellow). See also Fig. 6.

at λ_*, and we assume that the eigenvalue problem associated to the back has no nontrivial bounded solution at λ_*. For the proof, we refer the reader to [20], as well as [28], which introduced the approach.

7. Numerical results

In this section we illustrate our results numerically, utilizing the stripe and spot pattern of the cubic-quintic Swift–Hohenberg equation; note that spectral computations for these equations have been published previously in [3]. We confirm that localized roll solutions are stable if and only if the particular fronts and backs from which they are composed are stable, and further confirm that the number of unstable eigenvalues for the localized solution is equal to the sum of unstable eigenvalues for the underlying front and back.

Bifurcation diagrams were computed via numerical continuation in MATLAB, as described in §4 above. Operators were again computed using spectral differentiation matrices in the periodic y-direction, and centered finite differences in the x-direction, with $n_y = 8$ Fourier modes and $n_x = 800$ equidistant points, with boundary conditions and domains as indicated below. The built-in MATLAB routine eigs with initial shift sigma $= 0.5$ was used to compute the 20 largest eigenvalues of the operator linearized about each solution along a branch, with operator boundary conditions as indicated.

We first recall the bifurcation structure of these patterns, which was shown in Figs. 6 and 7, along with particular solution profiles. Fig. 9 shows an $\mathcal{R}$-symmetric branch, along with one cross-connecting and one self-connecting asymmetric branch. Note that the cross-connecting branch ends at a $\kappa\mathcal{R}$-symmetric branch (not shown). All information necessary for constructing its bifurcation structure is contained in the $\mathcal{R}$-symmetric branch.

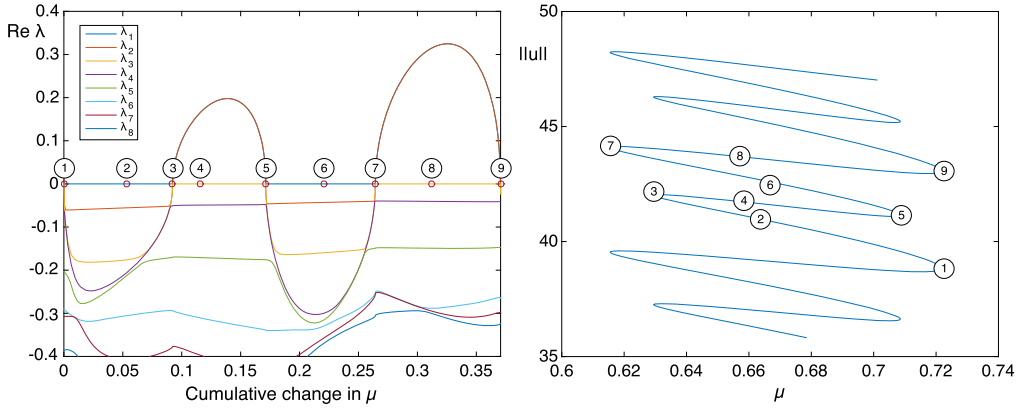


Fig. 10. Left: Eigenvalues for symmetric localized solutions along one period of the $\mathcal{R}$-symmetric snaking branch, with $\mathrm{Re}\,\lambda$ plotted as a function of the cumulative change in μ. Note there are two eigenvalues for the symmetric localized solution with $\lambda > 0$, corresponding to saddle node and pitchfork bifurcations, while the eigenvalue at $\lambda = 0$ is simple. Right: the bifurcation diagram for $\mathcal{R}$-symmetric localized solutions, with corresponding points on the eigenvalue diagram as indicated. Note that a point on the bifurcation diagram at right corresponds to a vertical slice in the eigenvalue plot at left.

7.1. Symmetric localized solutions

We begin by confirming that the multiplicity of eigenvalues is as expected. In Fig. 10 we show the eigenvalues computed along one period of the symmetric snaking branch, computed with Neumann boundary conditions on the domain $[-50, 50] \times [0, \pi]$. Along the x-axis we plot the cumulative change in the parameter μ, while the y-axis shows the real part of λ, which is in fact just λ in this case; all the eigenvalues are real as the operator $\mathcal{L}$ is self-adjoint.

In Fig. 11 we again show the eigenvalues for the symmetric localized solution, along with an overlay of the eigenvalues of a "front" solution computed on the domain $[0, 50] \times [0, \pi]$, again with Neumann boundary conditions. Of course, as soon as we begin computing on a bounded domain, we are effectively computing a localized solution. Our aim here is thus to illustrate that the correspondences and multiplicities are as expected, and we do not claim or attempt a test of exponential closeness. We indeed observe a single eigenvalue for the front with $\mathrm{Re}\,\lambda > 0$ giving rise to two such eigenvalues for the symmetric localized solution, which correspond to saddle node and pitchfork bifurcations. The eigenvalue of the symmetric localized solution at $\lambda = 0$ is simple. In Fig. 12, we focus on the eigenvalues with largest real part.

7.2. Asymmetric localized solutions

Focusing now on the asymmetric localized solutions, we consider both a cross-connecting and self-connecting branch, both of which were depicted in Fig. 9. We begin in Fig. 13 with eigenvalues of the cross-connecting branch of asymmetric localized solutions, with correspondences to the bifurcation diagram as indicated. In Fig. 14 we again show these eigenvalues, along with an overlay of the eigenvalues for the corresponding "front" and "back" solutions as described above. Here we find that one unstable eigenvalue arises from the front, and the other from the back, so that each is simple for the localized structure. In Fig. 15 we show a detail of Fig. 14, focusing on the leading eigenvalues. We observe that the number of eigenvalues with $\mathrm{Re}\,\lambda > 0$,

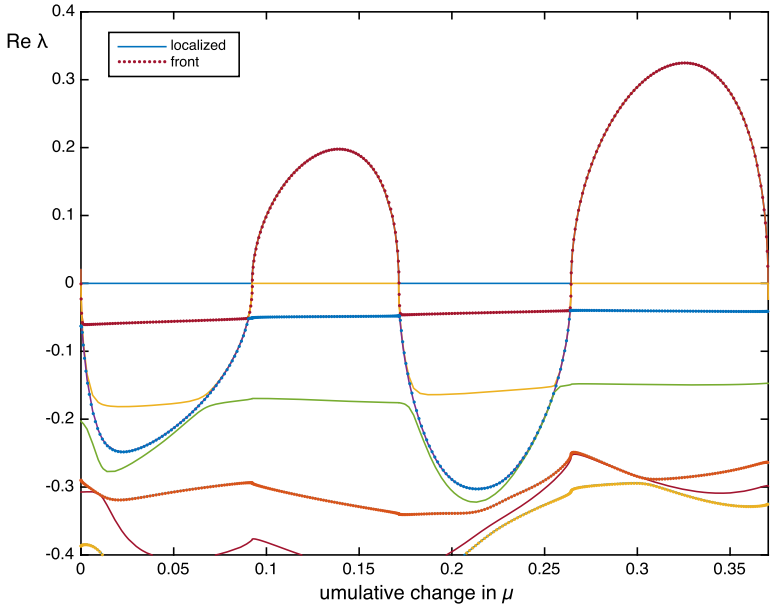


Fig. 11. Eigenvalues for symmetric localized solutions (solid), as shown in Fig. 10, along with eigenvalues for corresponding fronts (dots) overlaid along one period of a snaking branch.

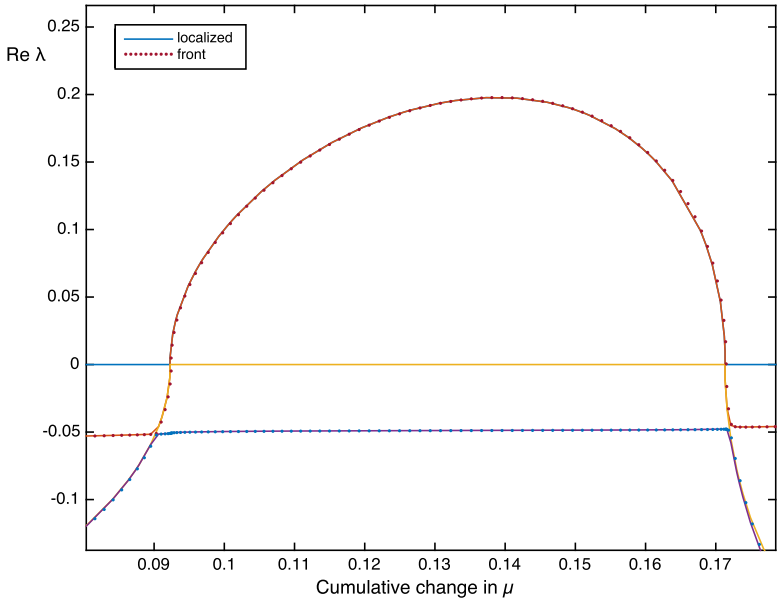


Fig. 12. Detail of the top left quadrant of Fig. 11, focusing on the leading eigenvalues for the first unstable portion of the symmetric branch.

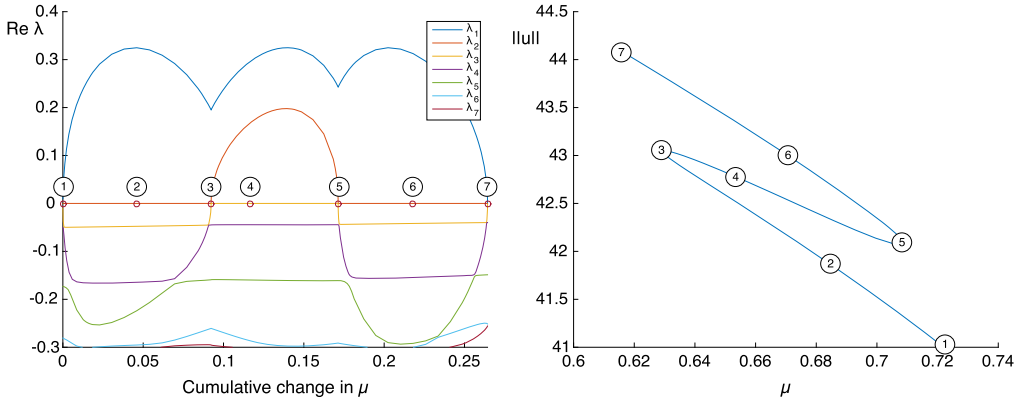


Fig. 13. Left: Eigenvalues for asymmetric localized solutions along a cross-connecting branch, with $\mathrm{Re}\,\lambda$ plotted as a function of the cumulative change in μ. Note that each of the eigenvalues with $\lambda > 0$ is simple, as is the eigenvalue at $\lambda = 0$. Right: the bifurcation diagram for the cross-connecting asymmetric localized solutions repeated from Fig. 9, with corresponding points on the eigenvalue diagram as indicated. Note that a point on the bifurcation diagram at right corresponds to a vertical slice in the eigenvalue plot at left.

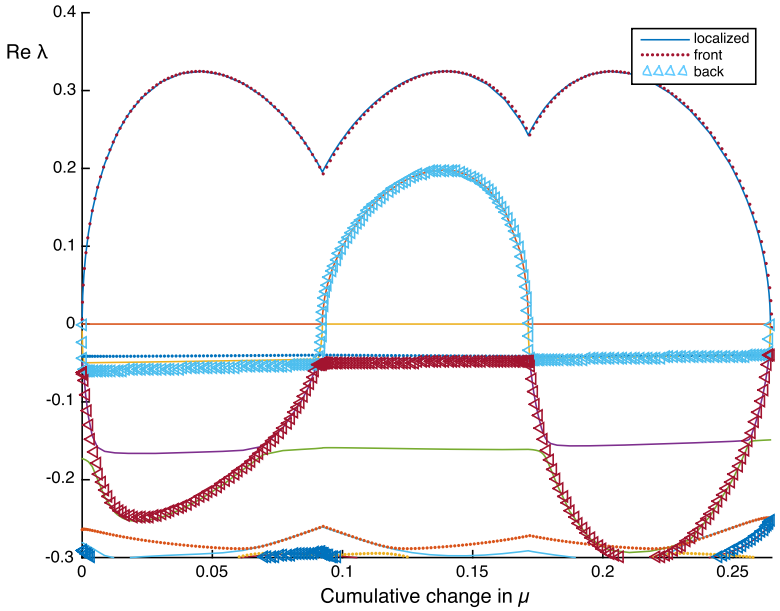


Fig. 14. Eigenvalues for asymmetric localized solutions along a cross-connecting branch (solid), with eigenvalues for corresponding fronts (dots) and backs (triangles) overlaid. Note that one unstable eigenvalue arises from the front, and the other from the back, so that each is simple.

changes from one to two and back to one, so that the localized solutions are unstable along the entire branch.

In Fig. 16 we show eigenvalues for the self-connecting branch of asymmetric localized solutions, again with correspondence to the bifurcation diagram as indicated. In Fig. 17 we repeat these eigenvalues, along with an overlay of the eigenvalues for the corresponding "front" and

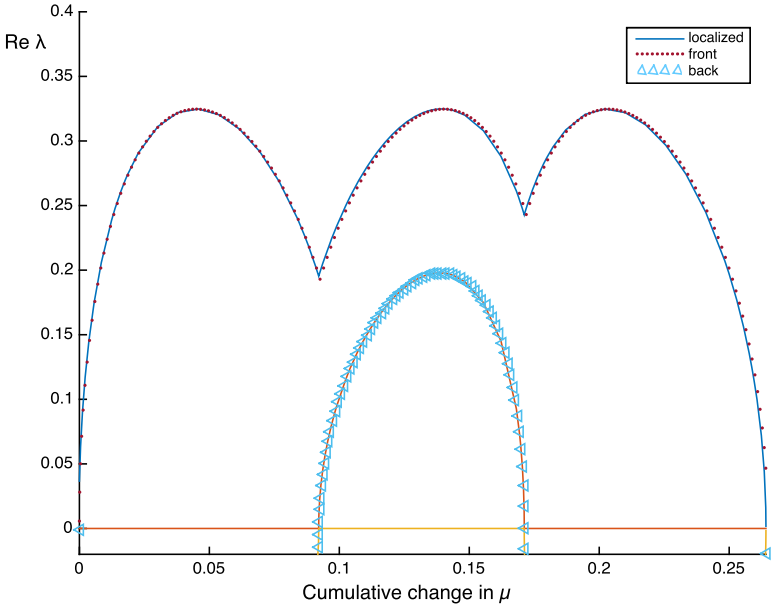


Fig. 15. Detail of Fig. 14, focusing on the leading eigenvalues.

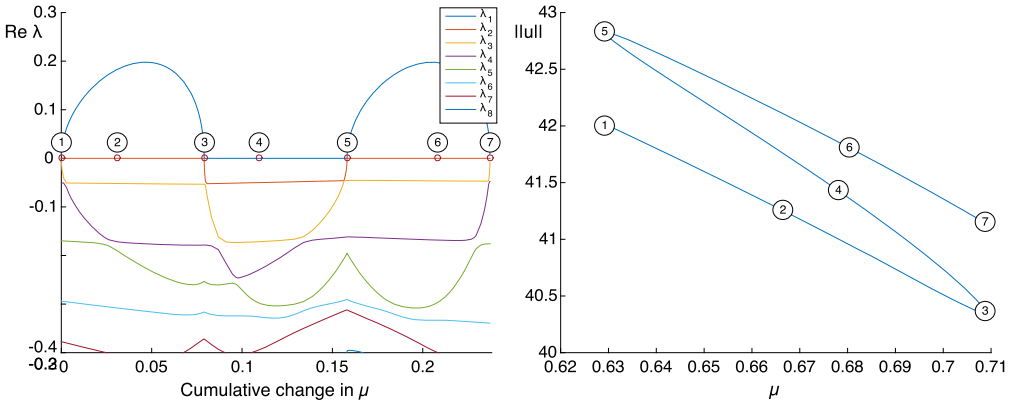


Fig. 16. Left: Eigenvalues for asymmetric localized solutions along a self-connecting branch, with $\mathrm{Re}\,\lambda$ plotted as a function of the cumulative change in μ. Note that each unstable eigenvalue is simple, as is the eigenvalue at $\lambda = 0$. Right: the bifurcation diagram for the cross-connecting asymmetric localized solutions repeated from Fig. 9, with corresponding points on the eigenvalue diagram as indicated. Note that a point on the bifurcation diagram at right corresponds to a vertical slice in the eigenvalue plot at left.

"back" solutions. We once again find that one unstable eigenvalue arises from the front and one from the back, so that both are simple. However, in contrast to the cross-connecting branch which has two unstable eigenvalues in the middle section, the middle section of the self-connecting branch has no unstable eigenvalues, as it arises from the combination of two stable front (or back) solutions. In Fig. 18 we show a detail of Fig. 14, focusing on the leading eigenvalues.

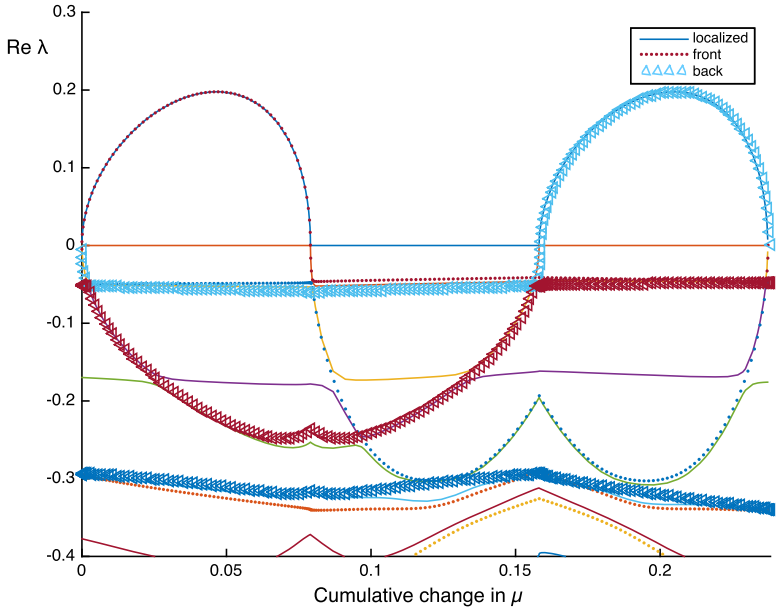


Fig. 17. Eigenvalues for asymmetric localized solutions along self-connecting branch (solid), with eigenvalues for corresponding fronts (dots) and backs (triangles) overlaid. Note that one unstable eigenvalue arises from the front, and the other from the back, so that each is simple. Both the front and the back are stable in the middle section, so that the asymmetric localized solution is as well for these parameter values.

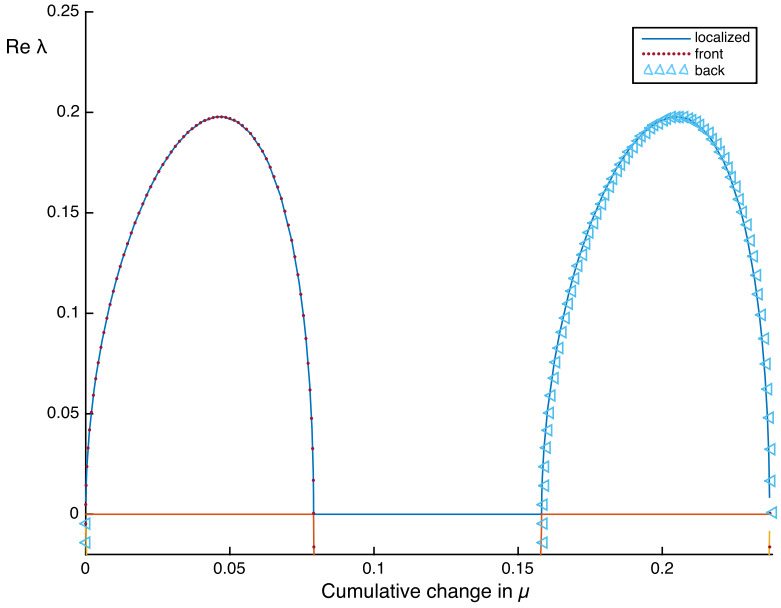


Fig. 18. Detail of Fig. 17, focusing on the leading eigenvalues.

8. Conclusion

In the preceding, we have rigorously shown that both symmetric and asymmetric solutions may be constructed by combining front and back solutions, and that the resulting localized solutions are exponentially close in the length L of the patterned region to the underlying fronts and backs. We have also illustrated this construction numerically. We then used this formulation to demonstrate that in the right half plane, eigenvalues of the localized solution are exponentially close in L to those of the front and back, and are added with multiplicity.

Consequently, if the underlying front and back have no eigenvalues in the open right half plane, the only possible scenario leading to eigenvalues of the localized solution in the open right half plane would be the creation of a pair eigenvalues near the origin, assuming that the front and back each have simple translation eigenvalues. This is the situation for monotone pulses, but as already mentioned, we use an extended Evans function in [22] to show it does not occur here. Taken together, these results show that if the front and back have no eigenvalues in the open right half plane, then neither do the localized solutions. Since there has sometimes been confusion in the literature, we note in particular that stable asymmetric solutions may therefore be formed from fronts and backs having no point spectrum in the right half plane, as seen in Figs. 16 and 17, for example.

As explored further in [22], there remain several interesting questions regarding the spectra of localized solutions, in particular the movement of saddle node and pitchfork eigenvalues in and out of the essential or absolute spectrum. Beyond this, analytical results on fully localized solutions in higher dimensions continue to be of interest, though at present it seems new techniques will be needed before progress in this direction can be achieved.

Acknowledgment

This work was supported by the National Science Foundation [DMS-1148284 and DMS-1408742].

Appendix A. Additional details for the construction of solutions in Theorem 1

In the following we explicitly construct solutions near the periodic orbits for $[-L, L]$ rather than invoking Lemma 2.7. In other words, we prove the first part of Lemma 2.7: there exist positive constants L_0 and η so that for all $L > L_0$ and $\varphi \in S^1$, there is a unique solution $u(x)$ of (2.1) defined for $x \in [-L, L]$ and lying in the zero-energy level set such that

$$u(-L) \in \Sigma_{in}, \quad u(L) \in \Sigma_{out}, \quad u^c(0) = \varphi,$$

and $u(x)$ stays near the periodic orbits for all $x \in [-L, L]$. We closely follow the proof of Lemma 6.2 in [4], which is an analogue of Lemma 2.7 for the Swift–Hohenberg equation posed as on ODE on $H^3_e(S^1) \times H^2_e(S^1) \times H^1_e(S^1) \times L^2_e(S^1)$ (the "e" stands for even periodic functions, but this is not essential). We believe these arguments bear repeating in order to clarify both the result and the pieces needed for extension to $\mathbb{R}^{2n}$ and appropriate Banach spaces. Moreover, the construction introduces techniques closely related to those which we employ in analyzing the stability of localized solutions.

The basic idea in the following is to construct two solutions, one of which starts in Σ_{in} and is the sum of a strong stable fiber to the periodic orbits plus an offset that is complementary in Σ_{in},

and a second of which ends in Σ_{out} and is the sum of a strong unstable fiber and an offset that is complementary in Σ_{out}. We then match these solutions in the middle. For convenience we write the two solutions as $u^+(x)$ for $x \in [0, L]$ and $u^-(x)$ for $x \in [-L, 0]$, so that $u(x) = u^+(x + L)$ for $x \in [-L, 0]$ and $u(-x) = u^-(x - L)$ for $x \in [0, L]$. The key ingredients necessary for the construction are smooth foliations and the existence of exponential dichotomies for the system when linearized about the strong (un)stable fibers.

Instead of using Fenichel coordinates in the three dimensional zero energy subspace, we proceed in the full four dimensional space. We have a two-parameter family of periodic orbits $\gamma(\cdot, \mu, e)$ which are smooth in μ, e, where $e := \mathcal{H}(\gamma(0, \mu, e), \mu)$ for e near zero, and we normalize the periodic orbits such that $\gamma(0, \mu, e) \in \operatorname{Fix} \mathcal{R}$ for each (μ, e). We assumed in Hypothesis 2.4 that $\alpha(\mu) > 0$ for all $\mu \in J$, where $\pm 2\pi\alpha(\mu)$ are the two Floquet exponents associated with the zero-energy periodic orbit. Thus there exists a neighborhood, E, of zero so that for all $(\mu, e) \in J \times E$, there exists an $\eta > 0$ such that the two Floquet exponents away from zero have magnitude greater than η, and the other two are fixed at zero.

Now for fixed (φ, μ, e), the strong stable fiber of $\gamma(x + \varphi, \mu, e)$, denoted by $q^+(x, \varphi, \mu, e)$, will be unique up to shifts, i.e., we need only fix its location at a single value of x to define it uniquely. We then use b^+ to parameterize q^+, so that $q^+(x, \varphi, \mu, e, b^+)$ is smooth in all arguments and there is a unique $b^+(\varphi, \mu, e)$ such that $q^+(0, \varphi, \mu, e, b^+(\varphi, \mu, e)) \in \Sigma_{in}$ and we can use b^+ to parameterize this condition.[1] We then have

$$\left|\gamma(x + \varphi, \mu, e) - q^+(x, \varphi, \mu, e, b^+)\right| \leq Ce^{-\eta x}, \quad x \geq 0, \tag{A.1}$$

for some $C > 0$. In the following C will denote any bounded uniform constant, without keeping track of these constants separately.

The strong unstable fiber $q^-(x, \varphi, \mu, e, b^-)$ of $\gamma(x + \varphi, \mu, e)$ is defined analogously and we have

$$\left|\gamma(x + \varphi, \mu, e) - q^-(x, \varphi, \mu, e, b^-)\right| \leq Ce^{\eta x}, \quad x \leq 0. \tag{A.2}$$

Now linearizing $u_x = f(u, \mu)$ about a particular fiber $q^+(x, \varphi^+, \mu, e^+, b^+)$ for $x \geq 0$ yields

$$u_x = f_u(q^+, \mu)u \tag{A.3}$$

and since the Floquet spectrum of the periodic orbits $\gamma(\cdot)$ has a gap between the two Floquet exponents fixed at 0 is bounded away from zero in absolute value by $\eta > 0$, (A.3) has exponential dichotomies (see, e.g., [29]) $\Phi^{cs}_+(x, y)$ and $\Phi^u_+(x, y)$ on $\mathbb{R}^+$ that are smooth in $(\varphi^+, \mu, e^+, b^+)$: fixing $\epsilon > 0$ such that $\eta > \epsilon > 0$, there exists a $C > 0$ such that

$$|\Phi^{cs}_+(x, y)| \leq Ce^{\epsilon(x-y)}, \quad x \geq y \geq 0, \tag{A.4}$$

$$|\Phi^u_+(x, y)| \leq Ce^{-\eta(y-x)}, \quad y \geq x \geq 0. \tag{A.5}$$

[1] In higher dimensional settings, in addition to capturing the x dependence, b^+ will parameterize the space corresponding to all but the weakest Floquet multiplier (see Hypothesis 3.1) and so will have dimension equal to the dimension of this space plus 1. We then have further smooth dependence of the solutions on these additional coordinates. See §3 for precise statements.

We denote by $P_+^{cs}(x)$ and $P_+^{u}(x)$ the projections associated with these exponential dichotomies, so that $P_+^{cs}(x) = \Phi_+^{cs}(x,x)$ and $P_+^{u}(x) = \Phi_+^{u}(x,x)$.

Similarly,

$$u_x = f_u(q^-, \mu)u \tag{A.6}$$

has exponential dichotomies $\Phi_-^{s}(x,y)$ and $\Phi_-^{cu}(x,y)$ on $\mathbb{R}^-$ that are smooth in $(\varphi^-, \mu, e^-, b^-)$, and we again denote by $P_-^{s}(x)$ and $P_-^{cu}(x)$ the projections associated with these dichotomies.

We wish to find solutions $u^{\pm}(x)$ to (2.1) that satisfy

$$\begin{aligned}
&u^+(x) = q^+(x, \varphi^+, \mu, e^+, b^+) + w^+(x), \quad x \in [0, L]\\
&u^+(0) \in \Sigma_{in}\\
&u^-(x) = q^-(x, \varphi^-, \mu, e^-, b^-) + w^-(x), \quad x \in [-L, 0]\\
&u^-(0) \in \Sigma_{out}\\
&u^+(L) = u^-(-L).
\end{aligned}$$

Since $q^+(x)$ solves (2.1), in order for $u^+(x)$ to satisfy (2.1) for $x \in [0, L]$ we must have

$$q_x^+ + w_x^+ = f(q^+ + w^+, \mu) = q_x^+ - f(q^+, \mu) + f(q^+ + w^+, \mu)$$

so that $w^+(x)$ satisfies

$$\begin{aligned}
w_x^+ &= f(q^+ + w^+, \mu) - f(q^+, \mu)\\
&= f_u(q^+, \mu)w^+ + N^+(q^+, w^+, \mu)
\end{aligned} \tag{A.7}$$

for $x \in [0, L]$ and where we have defined $N^+(q^+, w^+, \mu) := f(q^+ + w^+, \mu) - f(q^+, \mu) - f_u(q^+, \mu)w^+$. We observe that $N^+(q^+, w^+, \mu) = \mathrm{O}(|w^+|^2)$, and also note that if $f \in C^k$ then $N^+ \in C^{k-1}$ so that since q^+ is smooth in (φ, μ, e, b), N^+ is also smooth in each of these.

We have already noted the exponential dichotomies $\Phi_+^{cs}(x,y)$, $\Phi_+^{u}(x,y)$ for the linear part of (A.7), so using the variation of constants formula along with these dichotomies, and letting $a^+ \in \operatorname{Rg} P_+^{u}(L)$ an arbitrary element, we have

$$\begin{aligned}
w^+(x) &= \Phi_+^{u}(x, L)a^+ + \int_L^x \Phi_+^{u}(x,y)N^+(q^+(y), w^+(y), \mu)dy\\
&\quad + \int_0^x \Phi_+^{cs}(x,y)N^+(q^+(y), w^+(y), \mu)dy\\
&=: [\mathcal{F}(\mu, \varphi^+, e^+, b^+, a^+, w^+)](x)
\end{aligned} \tag{A.8}$$

so that solutions of (A.7) correspond to solutions of (A.8) on $[0, L]$. Since Φ_+^{cs} and Φ_+^{u} are smooth in $(\mu, \varphi^+, e^+, b^+)$, as is N^+, it is clear that $\mathcal{F}$ is smooth in $(\mu, \varphi^+, e^+, b^+, a^+)$. Now for each fixed $(\mu, \varphi^+, e^+, b^+, a^+)$, we show that $\mathcal{F}$ maps $C^0([0, L], \mathbb{R}^4)$ with weighted norm

$$\|w^+\| := \sup_{0\le x\le L} e^{\eta(L-x)}|w^+(x)| \tag{A.9}$$

into itself. Using the inequalities from (A.4) and (A.5) along with $N^+(q^+, w^+, \mu) = \mathrm{O}(|w^+|^2)$, we have:

$$\begin{aligned}
&\|\mathcal{F}(\mu, \varphi^+, e^+, b^+, a^+, w^+)\| \\
&\le C|a^+| + \sup_{0\le x\le L} Ce^{\eta(L-x)} \int_L^x e^{-\eta(y-x)} e^{-2\eta(L-y)} \|w^+\|^2 dy \\
&+ \sup_{0\le x\le L} Ke^{\eta(L-x)} \int_0^x e^{\epsilon(x-y)} e^{-2\eta(L-y)} \|w^+\|^2 dy \\
&= C|a^+| + \sup_{0\le x\le L} Ce^{-\eta L} \int_L^x e^{\eta y} \|w^+\|^2 dy + \sup_{0\le x\le L} Ce^{-\eta(L+x)} e^{\epsilon x} \int_0^x e^{(2\eta-\epsilon)y)} \|w^+\|^2 dy \\
&= C|a^+| + \sup_{0\le x\le L} Ce^{-\eta L} \left(e^{\eta x} - e^{\eta L}\right) \|w^+\|^2 + \sup_{0\le x\le L} Ke^{-\eta(L+x)} e^{\epsilon x} \left(e^{(2\eta-\epsilon)x} - 1\right) \|w^+\|^2 \\
&= C|a^+| + \sup_{0\le x\le L} C\left(e^{\eta(x-L)} - 1\right) \|w^+\|^2 + \sup_{0\le x\le L} Ce^{-\eta L} \left(e^{\eta x} - e^{(-\eta+\epsilon)x}\right) \|w^+\|^2 \\
&\le C|a^+| + C\|w^+\|^2,
\end{aligned}$$

where we have used the convention noted above of subsuming constants, independent of parameters and variables, into C. Then for any a^+ sufficiently small, (A.8) has a unique solution w^+ satisfying $|w^+(x)| \le C|a^+|e^{-\eta(L-x)}$ for $x \in [0, L]$ so that in particular $|w^+(0)| \le C|a^+|e^{-\eta L}$. Moreover, we have

$$w^+(L) - a^+ = \int_0^L \Phi_+^{cs}(L, y) N^+(q^+(y), w^+(y), \mu) dy$$

so that $|w^+(L) - a^+| \le C|a^+|^2$.

Similarly, $w^-(x)$ satisfies

$$\begin{aligned}
w^-(x) = \Phi_+^s(x, -L)a^- &+ \int_{-L}^x \Phi_-^s(x, y) N^-(q^-(y), w^-(y), \mu) dy \\
&+ \int_0^x \Phi_-^{cu}(x, y) N^-(q^-(y), w^-(y), \mu) dy
\end{aligned} \tag{A.10}$$

for $x \in [-L, 0]$, with $a- \in \mathrm{Rg}\, P_-^s(-L)$. Again for any a^- sufficiently small, (A.10) has a unique solution $w^-(x)$ satisfying $|w^-(x)| \le C|a^-|e^{-\eta(L-x)}$ for $x \in [0, L]$, and as before, $|w^-(-L) - a^-| \le C|a^-|^2$.

Then to satisfy the matching condition $u^+(L) = u^-(L)$, by definition we require $q^+(L, \varphi^+, \mu, e^+, b^+) + w^+(L) - q^-(-L, \varphi^-, \mu, e^-, b^-) - w^-(-L) = 0$, or, using (A.1) and (A.2),

$$\gamma(L + \varphi^+, \mu, e^+) - \gamma(-L + \varphi^-, \mu, e^-) + a^+ - a^- = \mathrm{O}(|a^+|^2 + |a^-|^2 + e^{-\eta L}). \tag{A.11}$$

We now fix (φ, e), and set

$$\begin{aligned} e^+ &= e + \hat{e}, \qquad & \varphi^+ &= \varphi + \hat{\varphi} - L \\ e^- &= e - \hat{e}, \qquad & \varphi^- &= \varphi - \hat{\varphi} + L \end{aligned}$$

for $(\hat{\varphi}, \hat{e})$ small. Then since for $(\hat{\varphi}, \hat{e}, a^+, a^-) = 0$ the left hand side of (A.11) vanishes, and its derivative with respect to these variables is invertible, we have a locally unique solution $(\hat{\varphi}_0, \hat{e}_0, a_0^+, a_0^-)$ of (A.11) that is smooth in (μ, e, φ, L), and we have $(\hat{\varphi}_0, \hat{e}_0, a_0^+, a_0^-) = \mathrm{O}(e^{-\eta L})$. We then set b^+, b^- such that $q^+(0, \varphi + \hat{\varphi}_0 - L, \mu, e + \hat{e}_0, b^+) \in \Sigma_{in}$ and $q^-(0, \varphi + \hat{\varphi}_0 - L, \mu, e + \hat{e}_0, b^+) \in \Sigma_{in}$. Then since $w^+(0) \in \operatorname{Rg} P_+^u(0)$ and $|w^+(0)| = \mathrm{O}(e^{-2\eta L})$, we have $u^+(0) \in \Sigma_{in}$. Similarly, $w^-(0) \in \operatorname{Rg} P_-^s(0)$ and $|w^-(0)| = \mathrm{O}(e^{-2\eta L})$, so that $u^-(0) \in \Sigma_{out}$.

We finish by establishing that for an appropriate choice of e, the solutions $u^\pm$ will lie in the zero-energy level set. We first show that $\mathcal{H}(u^+(0)) = \mathcal{H}(\gamma(0, \mu, e)) + \mathrm{O}(e^{-\eta L})$ so that $\frac{d}{de}\mathcal{H}(u^+(0)) \neq 0$ for e near 0 by assumption:

$$\begin{aligned} \mathcal{H}(u^+(0)) &= \mathcal{H}(u^+(L)) = \mathcal{H}(q^+(L, \mu, e, \varphi - L, b^+) + \mathrm{O}(e^{-\eta L})) \\ &= \mathcal{H}(\gamma(\varphi, \mu, e) + \mathrm{O}(e^{-\eta L})) = \mathcal{H}(\gamma(0, \mu, e)) + \mathrm{O}(e^{-\eta L}). \end{aligned}$$

Thus for e near 0, we can determine the unique $e = e(\mu, \varphi, L)$ so that $\mathcal{H}(u^+) = 0$.

References

[1] J.C. Alexander, R.A. Gardner, C.K.R.T. Jones, A topological invariant arising in the stability analysis of travelling waves, J. Reine Angew. Math. 410 (1990) 167–212.

[2] Y. Astrov, Y. Logvin, Formation of clusters of localized states in a gas discharge system via a self-completion scenario, Phys. Rev. Lett. 79 (16) (1997) 2983–2986.

[3] D. Avitabile, D.J.B. Lloyd, J. Burke, E. Knobloch, B. Sandstede, To snake or not to snake in the planar Swift–Hohenberg equation, SIAM J. Appl. Dyn. Syst. 9 (3) (2010) 704–733.

[4] M. Beck, J. Knobloch, D.J.B. Lloyd, B. Sandstede, T. Wagenknecht, Snakes, ladders, and isolas of localized patterns, SIAM J. Math. Anal. 41 (3) (2009) 936–972.

[5] S. Blanchflower, Magnetohydrodynamic convectons, Phys. A 261 (1–2) (1999) 74–81.

[6] U. Bortolozzo, M.G. Clerc, S. Residori, Solitary localized structures in a liquid crystal light-valve experiment, New J. Phys. 11 (2009) 093037.

[7] J. Burke, E. Knobloch, Localized states in the generalized Swift–Hohenberg equation, Phys. Rev. E 73 (5) (2006) 056211.

[8] M. Cross, P. Hohenberg, Pattern formation outside of equilibrium, Rev. Modern Phys. 65 (3) (1993) 851–1112.

[9] H. Dankowicz, F. Schilder, An extended continuation problem for bifurcation analysis in the presence of constraints, ASME J. Comput. Nonlinear Dyn. 6 (3) (2011) 031003.

[10] J.H.P. Dawes, The emergence of a coherent structure for coherent structures: localized states in nonlinear systems, Philos. Trans. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci. 368 (1924) (2010) 3519–3534.

[11] D. Henry, Geometric Theory of Semilinear Parabolic Equations, Lecture Notes in Mathematics, vol. 840, Springer-Verlag, 1981.

[12] G.W. Hunt, M.A. Peletier, A.R. Champneys, P.D. Woods, M.A. Wadee, C.J. Budd, G.J. Lord, Cellular buckling in long structures, Nonlinear Dynam. 21 (1) (2000) 3–29.

[13] T. Kapitula, B. Sandstede, Edge bifurcations for near integrable systems via Evans function techniques, SIAM J. Math. Anal. 33 (5) (2002) 1117–1143.
[14] T. Kapitula, B. Sandstede, Eigenvalues and resonances using the Evans function, Discrete Contin. Dyn. Syst. 10 (4) (2004) 857–869.
[15] K. Kirchgässner, Wave-solutions of reversible systems and applications, J. Differential Equations 45 (1982) 113–127.
[16] E. Knobloch, Spatially localized structures in dissipative systems: open problems, Nonlinearity 21 (2008) 45–60.
[17] E. Knobloch, Spatial localization in dissipative systems, Ann. Rev. Condens. Matter Phys. 6 (2015) 325–359.
[18] D.J.B. Lloyd, H. O'Farrell, On localised hotspots of an urban crime model, Phys. D 253 (2013) 23–39.
[19] D.J.B. Lloyd, B. Sandstede, D. Avitabile, A.R. Champneys, Localized hexagon patterns of the planar Swift–Hohenberg equation, SIAM J. Appl. Dyn. Syst. 7 (2008) 1049–1100.
[20] E. Makrides, Existence and Stability of Spatially Localized Planar Patterns, Ph.D. thesis, Brown University, 2016.
[21] E. Makrides, B. Sandstede, Predicting the bifurcation structure of localized snaking patterns, Phys. D 268 (2014) 59–78.
[22] E. Makrides, B. Sandstede, Spatially localized patterns and eigenvalues in the essential, or absolute spectrum, unpublished, 2018.
[23] J.M. McSloy, W.J. Firth, G.K. Harkness, G.L. Oppo, Computationally determined existence and stability of transverse structures II. Multipeaked cavity solitons, Phys. Rev. E 66 (2002) 046606.
[24] E. Meron, Pattern-formation approach to modelling spatially extended ecosystems, Ecol. Model. 234 (2012) 70–82.
[25] S. Nii, An extension of the stability index for traveling-wave solutions and its application to bifurcations, SIAM J. Math. Anal. 28 (2) (1997) 402–433.
[26] D. Peterhof, B. Sandstede, A. Scheel, Exponential dichotomies for solitary-wave solutions of semilinear elliptic equations on infinite cylinders, J. Differential Equations 140 (2) (1997) 266–308.
[27] S. Pirkl, P. Ribière, P. Oswald, Forming process and stability of bubble domains in dielectrically positive cholesteric liquid crystals, Liq. Cryst. 13 (1993) 413–415.
[28] B. Sandstede, Stability of multiple-pulse solutions, Trans. Amer. Math. Soc. 350 (2) (1998) 429–472.
[29] B. Sandstede, Stability of traveling waves, in: B. Fiedler (Ed.), Handbook of Dynamical Systems II, Elsevier, 2004, pp. 983–1055.
[30] B. Sandstede, A. Scheel, Absolute and convective instabilities of waves on unbounded and large bounded domains, Phys. D 145 (2000) 233–277.
[31] B. Sandstede, A. Scheel, Defects in oscillatory media: toward a classification, SIAM J. Appl. Dyn. Syst. 3 (1) (2004) 1–68.
[32] T.M. Schneider, J.F. Gibson, J. Burke, Snakes and ladders: localized solutions of plane Couette flow, Phys. Rev. Lett. 104 (2010) 104501.
[33] M.B. Short, M.R. D'Orsogna, V.B. Pasour, G.E. Tita, P.J. Brantingham, A.L. Bertozzi, L.B. Chayes, A statistical model of criminal behavior, Math. Models Methods Appl. Sci. 18 (2008) 1249–1267.